Bond University

Research Repository

Do corruption, income inequality and redistribution hasten transition towards (non)renewable energy economy?

Acheampong, Alex O.; Boateng, Elliot; Annor, Collins Baah

Published in:
Structural Change and Economic Dynamics

DOI:
10.1016/j.strueco.2023.11.006

Licence:
CC BY

Link to output in Bond University research repository.

Recommended citation(APA):
Acheampong, A. O., Boateng, E., & Annor, C. B. (2024). Do corruption, income inequality and redistribution hasten transition towards (non)renewable energy economy? *Structural Change and Economic Dynamics*, *68*, 329-354. https://doi.org/10.1016/j.strueco.2023.11.006

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

For more information, or if you believe that this document breaches copyright, please contact the Bond University research repository coordinator.

Download date: 12 Dec 2025

Structural Change and Economic Dynamics 68 (2024) 329–354

Contents lists available at ScienceDirect

Structural Change and Economic Dynamics

journal homepage: www.elsevier.com/locate/strueco

Do corruption, income inequality and redistribution hasten transition towards (non)renewable energy economy?

Alex O. Acheampong [a,b,*], Elliot Boateng [c,d], Collins Baah Annor [e]

[a] Bond Business School, Bond University, Gold Coast, Australia
[b] Centre for Data Analytics, Bond University, Gold Coast, Australia
[c] Department of Economics, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
[d] Centre for African Research, Engagement and Partnerships (CARE-P), University of Newcastle, Australia
[e] Policy, Planning, Monitoring and Evaluation Department (PPMED), Ghana Health Service, Goaso Municipal Hospital, Goaso, Ghana

ARTICLE INFO

JEL classification:
C24
O13

Keywords:
Energy transition
Net-zero
Income redistribution
Corruption
Political economy

ABSTRACT

This study examines the impact of corruption, income inequality, and redistribution on energy consumption (renewable, nonrenewable, and total) in 166 countries. Panel data and the two-step dynamic system-generalized method of moments techniques are utilised. Our findings show that net (post-tax/transfer Gini index) and market (pre-tax/transfer Gini index) income inequality reduce total energy and renewable energy consumption while increasing nonrenewable energy consumption. The results indicate that income redistribution drives total energy and renewable energy consumption while decreasing nonrenewable energy use. The study highlights that control of corruption increases total energy, renewable, and nonrenewable energy consumption. Moderation and marginal effect analysis suggest that when control of corruption increases, income inequality increases total energy, nonrenewable, and renewable energy consumption. Again, when corruption is effectively controlled, income redistribution increases renewable energy consumption while reducing nonrenewable energy consumption. These findings differ amongst regions. Following these findings, we recommend that policymakers should focus on reducing corruption and addressing income inequality through redistribution policies to accelerate the transition towards a renewable energy economy.

1. Introduction

This study contributes to policy discussions on the transition to a zero-carbon economy, focusing on corruption, income inequality, and redistribution's role in the global and regional energy transition. The rise in global greenhouse gases (GHGs) and specifically, carbon emissions , driven by activities like agriculture, industry, and fossil fuel combustion, poses a significant challenges (Acheampong, Boateng & Amponsah, 2022; Sharma, 2011; Jin and Kim, 2018; Wang et al., 2020). In line with the Paris Agreement, Sustainable Development Goals (SDGs), UN agreements on Disaster Risk Reduction, and COP26, global stakeholders pledged to mitigate climate change (Mountford et al., 2021; Pörtner et al., 2022). To achieve this, expediting the global energy transition is crucial, and the Glasgow Climate Pact emphasizes the need to reduce nonrenewable energy (coal) and increase the share of renewable energy in the global energy mix to limit global warming to 1.5 °C. In view of this, research on achieving a renewable energy economy and the pathway to net zero has gained momentum recently (Dogan et al., 2022; Dogan and Seker, 2016a, 2016b; Inglesi-Lotz and Dogan, 2018; Xu et al., 2022; Yang et al., 2022). Policymakers need to comprehend the factors influencing renewable and nonrenewable energy consumption to promote the transition and enhance energy efficiency. Several studies have investigated factors such as income, oil price, carbon emissions, trade openness, financial development, foreign direct investment, human capital, governance, and the Kyoto Protocol's impact on renewable, nonrenewable, or total energy consumption (e.g., Acheampong et al., 2021; Amuakwa-Mensah et al., 2018; Bataille and Melton, 2017; Doytch and Narayan, 2016; Liu, Zhou, Huang, and Hao, 2018; Rafiq et al., 2016; Troster et al., 2018).

Despite the significant research into renewable, nonrenewable, and total energy consumption and its implication for emissions reduction, three salient issues remain unresolved in the energy transition, net zero and green economic transformation debate: (1) the role of income inequality, (2) the impact of income redistribution, and (3) the

* Corresponding author.
E-mail address: aacheamp@bond.edu.au (A.O. Acheampong).

https://doi.org/10.1016/j.strueco.2023.11.006
Received 19 February 2023; Received in revised form 4 October 2023; Accepted 11 November 2023
Available online 13 November 2023
0954-349X/© 2023 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

fundamental role of corruption. A critical review of existing studies shows a general lack of comprehensive studies on income inequalities, redistribution, and renewable and nonrenewable energy consumption dynamics. Closely related studies have reported inconclusive results (Acheampong et al., 2022; Liu et al., 2018; Uzar and Eyuboglu, 2019; Churchill et al., 2021). Hao et al. (2016) and Borghesi (2006) argued that income inequality explains the cross-country differences in renewable and nonrenewable energy consumption. Income inequality undermines collective actions to transition towards a renewable energy economy and mitigate environmental pollution. Income inequality worsens societies' vulnerability to weak social cohesion and mistrust, thus making people less sensitive to environmental protection (see Churchill et al., 2021; Asongu and Odhiambo, 2021; Berthe and Elie, 2015; Wilkinson and Pickett, 2010) and increased social status competition, which widens the gap between renewable and nonrenewable usage by the households (Berthe and Elie, 2015; Wilkinson and Pickett, 2010). Also, given that nonrenewable energy is less expensive than renewable energy, households with less income would depend heavily on nonrenewable energy usage at the expense of renewable energy usage. Therefore, increasing income inequality between the rich and the poor would minimise renewable energy usage and promote nonrenewable energy consumption. Income inequality, thus, creates power asymmetry between the poor and the rich (Boyce, 1994). With power asymmetry, wealthy people can render renewable energy policies ineffective and reduce renewable energy usage while increasing nonrenewable energy consumption (Strunz, Gawel, and Lehmann, 2016; Uzar, 2020). Uzar (2020) indicates that the wealthy (elite) determine the policy process in an unequal society, and they can prevent incentives and subsidies to boost renewable energy usage and investment. The discussions show that income inequality is a bane to energy transition and that effective policies that support income redistribution would help to achieve the energy transition agenda.

Income redistribution policies are paramount in addressing income inequality (Goñi et al., 2011; Milanovic, 2000). Income redistribution plays a crucial role in energy transition, but its effect on energy consumption has been understudied in the literature. Transitioning towards a renewable energy economy and improving energy efficiency depends on household purchasing ability, consumer behaviour and the affordability of renewable and nonrenewable energy. Poorer households tend to patronise non-renewable energy, such as fossil energy, because of their affordability relative to renewable energy. Therefore, we argue that increasing income equality through income redistribution could enable low-income households to consume renewable energy while improving energy efficiency simultaneously. For instance, a recent study by Sampedro et al. (2022) showed that when income distribution becomes more egalitarian in the United States (US), residential energy consumption could be 10% higher in 2100. The author further noted that residential energy consumption could be 19% lower when there is less equitable income distribution within the US. Although Sampedro et al. (2022) study highlights the importance of income redistribution on energy demand, it does not offer insight if the impact of income redistribution differs between renewable and nonrenewable energy. This presents a significant literature gap that needs to be addressed.

Besides, evidence shows that promoting renewable energy usage while phasing out nonrenewable use to create a pathway for a green environment and sustainability requires institutional support from national, sub-national, and international actors (Blau, 2017; Boyle, 2018; Nair et al., 2021). However, many countries face institutional challenges in emissions-mitigating strategies and renewable energy policies. Increased corruption constrains the design, implementation, and enforcement of policies needed to ensure energy transition and environmental sustainability (see Akhbari and Nejati, 2019; Leitão, 2010; Lisciandra and Migliardo, 2017). Corruption contributes to ineffective policies and practices, weak enforcement and implementation of policies, and the embezzlement of funds meant to promote green environmental programs (Biswas et al., 2012; Lopez and Mitra, 2000) and could hinder efforts toward renewable energy transition. Therefore, addressing corruption practices is crucial, as it enables governments to design and implement sound energy policies vital to promoting energy efficiency and adopting renewable energy (Bhattacharya et al., 2017). Notwithstanding, the unresolved question is whether corruption plays a significant role in the impact of income inequality and redistribution on renewable and nonrenewable energy consumption.

In this study, we contribute to the literature by investigating the impact of corruption, income inequality, and redistribution on renewable, nonrenewable, and total energy consumption. Examining the role of corruption, income inequality, and redistribution on renewable, nonrenewable, and total energy consumption has important implications for achieving the energy transition and net-zero agenda. For instance, researchers (e.g., Sadorsky, 2010, 2011a) argue that understanding the factors that influence energy consumption (including both renewable and nonrenewable) is an essential area of research since they shape energy planning and environmental conservation. First, understanding the role of corruption, income inequality, and redistribution on renewable, nonrenewable, and total energy consumption is vital for providing a futuristic understanding of how energy consumption will change to avoid under-estimation or over-estimation of actual energy demand. Second, understanding the role of corruption, income inequality, and redistribution on renewable, nonrenewable, and total energy consumption modelling is critical to managing GHGs and achieving the net zero, which was at the centre of the COP26 discussions; given its relevant in curtailing GHGs concentrations in the atmosphere. Following this study, we seek to provide answers to the following questions:

1. *Do corruption, income inequality, and redistribution contribute to over-dependence on nonrenewable energy use?*
2. *Do corruption, income inequality, and redistribution hinder the transition to a renewable energy economy?*
3. *Do corruption, income inequality, and redistribution explain the variation in (non)renewable energy consumption between geographical regions?*

Our study's contribution, novelty, and uniqueness to the literature on energy transition are discussed as follows: First, unlike previous studies with smaller sample sizes, we utilise a comprehensive dataset comprising 166 countries. This larger sample provides more reliable and valid conclusions regarding the impact of corruption, income inequality, and income redistribution on renewable, nonrenewable, and total energy consumption. While previous studies (e.g., Awaworyi Churchill et al., 2021; Asongu and Odhiambo, 2021; Uzar, 2020) have explored the effect of income inequality on renewable energy consumption, our study is the first, to the best of our knowledge, to provide empirical evidence on the contribution of relative and absolute income redistribution to different energy consumption variables. Most importantly, we test if net (that is, post-tax/post-transfer Gini) income inequality and market (pre-tax/pre-transfer Gini) income inequality have a disparate effect on renewable and nonrenewable energy as well as total energy consumption. We show that net (post-tax/transfer Gini index) and market (pre-tax/transfer Gini index) income inequality reduces total energy and renewable energy consumption while increasing nonrenewable energy consumption. Our study also shows that absolute and relative income redistribution drives total energy and renewable energy consumption while lessening nonrenewable energy use. By providing these contributions, our study enhances the understanding of the relationship between corruption, income inequality, income redistribution, and energy consumption, thereby offering valuable insights for policymakers and researchers in the field of energy transition.

Second, earlier studies on income inequality and energy consumption were silent on corruption's role in the income inequality-energy consumption nexus (see, for instance, Asongu and Odhiambo, 2021; Awaworyi Churchill et al., 2021). We argue that such omission implies researchers attach less weight to the role of institutional settings in the

income inequality-energy consumption nexus. By extending existing literature to account for the institutional environment (control of corruption) in the energy consumption models, we show that the control of corruption conditions the impact of income inequality and redistribution on the energy variables. Thus, our findings show that control of corruption increased total energy, renewable, and nonrenewable energy consumption. Also, moderation and marginal effect analysis showed that with effective corruption control of corruption, the post-tax/transfer Gini index increases total energy and nonrenewable energy consumption while reducing renewable energy consumption. We also noted from the moderation and marginal effect results that when control of corruption increases, the pre-tax/transfer Gini index increases total energy, nonrenewable, and renewable energy consumption. At the same time, moderation and marginal effect analysis revealed that with control of corruption, both relative and absolute income redistribution increases renewable energy consumption while reducing nonrenewable energy consumption. By considering the role of corruption, our study sheds light on the interplay between income inequality, redistribution, and energy consumption, highlighting the importance of institutional factors in shaping energy transition dynamics.

Third, our study contributes to knowledge by examining the role of corruption, income inequality, and redistribution in explaining variations in renewable, nonrenewable, and energy-intensity consumption across different geographical regions. We conduct a disaggregated analysis at various regional levels to capture regional heterogeneity, expanding our understanding of the relationship between corruption, income inequality, redistribution, and energy consumption. Our findings indicate that corruption, income inequality, and redistribution are significant factors accounting for variations in renewable and nonrenewable energy consumption and energy intensity amongst different regions. From a policy perspective, our research provides valuable insights for policy discussions, particularly in COP26, to develop strategies for promoting renewable energy use and phasing out nonrenewable energy. While financial resource mobilization has been a central focus in achieving net-zero during COP26, our study highlights the importance of considering corruption, income inequality, and redistribution as crucial concerns for policymakers aiming to achieve net-zero through the transition to renewable energy. Based on our results, we recommend that achieving the objectives of COP26, such as net-zero and energy transition, requires effective income redistribution and a strong commitment from policymakers to address corruption. By emphasizing these recommendations, our study contributes to the ongoing dialogue on energy transition and provides valuable guidance for policymakers striving to achieve sustainable and equitable renewable energy goals.

The rest of the paper is organised as follows. Literature review and hypothesis formulations are discussed in Section 2, and methodology and data are presented in Section 3. Section 4 presents the empirical analysis and discussion, and Section 5 provides our findings' conclusion and policy implications.

2. Review of related literature review

2.1. Income inequality, redistribution, and (non)renewable energy consumption

There exists no consensus within the theoretical literature on how income inequality relates to renewable energy or the direction of causality between them. The reason for such disagreements is that hitherto, myriad theoretical arguments have been expounded to explain the relationship, especially on how income inequality relates to environmental degradation. Several channels exist through which income inequality impacts (non)renewable energy consumption.

Widening income inequality weakens societal cohesion by disrupting harmony and trust amongst individuals, eventually decreasing confidence and weakening society (Berthe and Elie, 2015; Wilkinson and Pickett, 2010; Boyce, 2007). Disrupting harmonious existence amongst individuals represents the destruction of collective effort and promotes individualism, which means both the wealthy and the poor will decide to operate individually regarding environmental use (Uzar, 2020). Boyce (1994) iterates that the absence of environmental sensitivity in societies is characterised by a lack of social cohesion and collective action resulting from unequal income distribution. The response of people experiencing poverty to deepening unfair income distribution is to engage the environment extensively to satisfy their basic needs (Wolde-Rufael and Idowu, 2017). Unfortunately, the engagement of the environment by people with low incomes to satisfy their basic needs leads to the deterioration of the environment. For instance, Laurent (2015) showed considerable forest deterioration is observed in Madagascar and Haiti whenever people with low incomes use the forest to sustain their lives. This behaviour is because they give little attention to the long-run consequences of their present consumption decisions (Uzar, 2020) in favour of the short-run benefits (Berthe and Elie, 2015). Societies characterised by income inequality are identified by conspicuous consumption and competition for status to maintain or improve their status in society (Berthe and Elie, 2015). Those who do not have enough income tend to imitate the consumption pattern of the wealthy class (Uzar, 2020). Under instances where the consumption pattern does not favour clean energy usage, they place pressure on the environment. Thus, Consumerism, although not a function of environmental quality, contributes significantly to adopting renewable energy through the desire to attain a certain societal status (Berthe and Elie, 2015). Wilkinson and Pickett (2010) supported this when they iterated that rivalling for social status will increase within societies where income inequality is widened.

Another channel establishing the linkage between income inequality, redistribution, and the consumption of (non)renewable energy is economic factors. Regarding the economic channel, income inequalities influence the consumption of (non)renewable through their effect on social norms. Political factors constitute another channel in establishing the linkages between income inequality, redistribution, and (non)renewable energy. Regardless of how strong environmental policies are, they are unlikely to serve the interests of all persons. Furthermore, the political demands of some of these social groups to whom the environmental policies did not favour considerably affect designing environmental policies (Boyce, 1994). As a result, the power relationship amongst various social groups constitutes a key determinant in adopting (non) renewable energy.

Sequeira and Santos (2018) added to this discussion by indicating the capacity and structures of state institutions can play a crucial role in advancing the quest for environmental protection and renewable energy consumption. Institutions are essential in energy use, income distribution, and environmental policies (Cadoret and Padovano, 2016). Income inequality may reflect weak institutions (Uzar, 2020), which could distort the efficiency of investments (Mehrara et al., 2015) and weaken the effects of environmental policies. Individuals with high incomes may leverage these weak institutions and engage in activities detrimental to the environment. Inefficiencies in investment imply inefficiencies in renewable energy investments, which thus affects the decision to adopt (non) renewable energy. Consequently, the extent of power inequalities emanating from income inequality within societies influences the effectiveness of institutions and the consumption of (non)renewable energy. Another viewpoint lies in the rent-seeking attitude of poor groups to maximise profits and seek to accrue benefits from renewable energy consumption (Strunz et al., 2016). Another potential argument is that high-income groups eventually become more concerned about post-materialist issues, including the environment (Scruggs, 1998), affirming the theoretical position of the Environmental Kuznets curve.

Empirical evidence in favour of this strand of argument is observed in Hao et al. (2016) and Borghesi (2006). By contrast, there exists the inequality hypothesis within the theoretical literature that argues that a trade-off exists between the quality of the environment and redistributive policies, which implies that an inverse association exists between

income inequality and environmental degradation with empirical evidence of this argument found in Heerink et al. (2001). In line with the two arguments discussed above, it could be hypothesised that a positive nexus is observed between income inequality and environmental degradation, representing the equality hypothesis. The second hypothesis represents an association between income inequality and environmental degradation, reflecting the inequality hypothesis. In line with the focus of the study, it is argued that in the case of the first hypothesis, an increase in income inequality would lead to environmental degradation, the effect of which would be a reduction (increase) in the consumption of renewable (nonrenewable) energy. Thus, the first hypothesis posits a negative (positive) association between income inequality and renewable (nonrenewable) energy consumption. In the case of the second hypothesis, it is argued that for every increase in income inequality, there will be a reduction in environmental degradation, which would lead to an increase (decrease) in renewable energy consumption, which means a positive (negative) association exist between income inequality and renewable (nonrenewable) energy consumption. In line with Grunewald et al. (2017), the theoretical literature on the relationship between income inequality and environmental degradation or income inequality and (non)renewable energy remains inconclusive, especially given that the association may also be non-monotonous.

For instance, Churchill et al. (2021) examined the role of income inequality on renewable energy consumption and their disaggregated form for seventeen countries. With the panel data spanning 1990 to 2016, the study engaged both parametric and non-parametric local linear dummy variable estimation and observed that the effect of income inequality on renewable energy in the short run was negative up to 2010 when the relationship became positive. The renewable energy measure also allowed for observing heterogeneous impacts of income inequality on renewable energy consumption. The time-varying impacts show that income inequality affects renewable energy consumption in several ways, with some mediums dominating over others at specific periods. Uzar (2020) investigated, theoretically and empirically, the effect of income inequality on renewable energy consumption for 43 developed and developing countries from 2000 to 2015, with indicators covering economic, institutional, and environmental factors that influence renewable energy consumption. The methodology engaged included the panel Autoregressive Distributed Lag Model and Pooled Mean Group approach, and the finding was that an improvement in income inequality would promote renewable energy consumption. Controlled corruption and increased carbon dioxide emissions lead to a rise in renewable energy consumption. Shahbaz et al. (2022) assessed the nexus between fiscal decentralisation and demand for renewable energy using quarterly frequency data in the case of China for the study period 1980 to 2018. The Autoregressive Distributed Lag model was used for the estimation. The results showed that while fiscal decentralisation helps to increase demand for renewable energy, inequalities in income lead to reductions in renewable energy consumption. Other factors that cause increases in renewable energy consumption include economic growth, urbanisation, and economic globalisation.

Bai et al. (2020) sought to investigate the dynamics between income inequality, renewable energy technology innovation, and carbon-dioxide emissions using provincial panel data from China covering the period 2000 to 2015. The essential methods used for the study included the panel fixed effect model and a panel threshold model to assess the nonlinearity of the relationship between these variables. The studies showed that renewable energy technological innovation could reduce per capita carbon dioxide emissions. However, an increase in income inequality will hinder the abatement impact of renewable energy technological innovations on per capita emissions. This widening income inequality could eventually cause renewable energy technology innovation to worsen per capita emissions instead of mitigating them. The panel threshold model reveals that the effect of renewable energy technological innovation on per capita emissions has single-threshold repercussions in terms of income inequality. Income inequalities lower than the threshold value would imply that the effect of renewable energy technological innovation on per capita carbon emissions is insignificant. However, income inequalities above the threshold and within the higher income inequality threshold, renewable energy technology innovation positively impacts per capita carbon emissions.

To answer the question of how income inequality influences energy efficiency, Liu et al. (2020), in response to serious concerns experienced in the Belt and Road Initiative countries, explored panel data of 33 Belt and Road Initiative countries with data spanning from 2000 to 2016 with the aid of the dynamic Data Envelopment Analysis model and the Tobit regression model. The findings showed that energy efficiency within the selected countries was generally low over the entire study period. On the other hand, the energy efficiency of high-income Belt and Road Initiative countries was observed to be relatively higher than that of the middle- and low-income Belt and Road countries. Furthermore, while the energy efficiency of the studied countries has fallen since 2013, there was a noticeable u-shaped impact of income inequality on energy efficiency. The effect of income inequality on energy efficiency is heterogeneous for economies with varying income levels with observable U-shaped and inverted U-shaped relations for high-income Belt and Road Initiative countries and middle-and low-income Belt and Road Initiative countries, respectively. Economic growth, structure, and industrialisation are other factors affecting energy efficiency.

Tan and Uprasen (2021) undertook a study on the asymmetric effect of income inequality on renewable energy consumption amongst the Association of Southeast Asian Nations (ASEAN) for the period 1990 to 2015. Four main measures of income inequality were employed on a nonlinear panel autoregressive distributed lag (ARDL) model. The authors showed that by reducing income inequalities, the demand and consumption of renewable energy increase in the long run. The study observed an asymmetric effect and shows that widening income inequality considerably affects renewable energy consumption relative to when income inequality improves. A bidirectional causality was found between worsened income inequality and renewable energy consumption. Asongu and Odhiambo (2021) researched the linkages between financial development, renewable energy consumption, and income inequality within 39 Sub-Saharan African countries using the Generalised Method of Moments and the Quantile Regression with panel data covering 2004 to 2014. Three income inequality indicators, the Gini coefficient, the Atkinson index, and the Palma ratio, were used for the study. The results show that whereas financial development encourages renewable energy consumption, income inequality hinders the underlying positive impacts. The results from the Quantile Regression show that the Generalised Method of Moments can only withstand empirical validity in the bottom quantiles of the distribution of renewable energy consumption. In Table 1, Panel A, we summarise the empirical studies on the effect of income inequality and redistribution on primary and (non)renewable energy consumption. From the theoretical and empirical literature, we test the following hypothesis:

2.2. Corruption, income inequality and (non)renewable energy consumption

The empirical literature has shown that corruption widens income inequalities (Adams and Klobodu, 2016; Sulemana and Kpienbaareh, 2018). High levels of corruption promote a rise in unemployment rates in an economy. This outcome breeds income inequality, which further makes poorer the economically vulnerable in society. Furthermore, widening income inequality increases household debt, which deepens income inequality (Berisha et al., 2018), indicating that income inequality for persons living in highly corrupt countries is a vicious cycle. The effect of the above is that growth is impeded as Gyimah-Brempong (2002) demonstrated that corruption increases inequalities in income and, therefore, a combined impact of declines in economic growth and widening income inequality further places a higher burden on the poor due to corruption. Bakre et al. (2017) show

Table 1
Summary of related studies on corruption, income inequality, redistribution and energy consumption.

Author & year	Title / Objective	Sample	Results / findings
Panel (A) Income inequality, redistribution, and (non)renewable energy consumption			
Heerink et al. (2001).	Examine income inequality and the environment: aggregation bias in environmental Kuznets curves	Cross-national regression analyses; the Shafik and Bandyopadhyay (1992) data.	The findings shows that a trade-off exists between the quality of the environment and redistributive policies, implying that an inverse association exists between income inequality and environmental degradation.
Churchill et al. (2021),	The role of income inequality on renewable energy consumption and their disaggregated form for seventeen countries	A panel of 17 nations over the period 1990 to 2016; time-varying evolution using non-parametric techniques.	They observed that the effect of income inequality on renewable energy in the short run was negative up to 2010 when the relationship became positive. The time-varying effects reflect that income inequality influences renewable energy consumption via various channels, and at specific times, some channels appear to be more dominant.
Uzar (2020)	Is income inequality a driver for renewable energy consumption?	43 developed and developing countries from 2000 to 2015; the panel ARDL method	Fairer income distribution positively affects renewable energy consumption.
Shahbaz et al. (2022)	Fiscal decentralization as new determinant of renewable energy demand in China: The role of income inequality and urbanization.	Quarterly frequency data in the case of China for the study period 1980 to 2018	The results showed that while fiscal decentralization helps to increase demand for renewable energy, inequalities in income lead to reductions in renewable energy consumption
Bai et al. (2020)	Will income inequality influence the abatement effect of renewable energy technological innovation on carbon dioxide emissions?	Provincial panel data from China covering the period 2000 to 2015	The studies showed that renewable energy technological innovation could reduce per capita carbon dioxide emissions. However, an increase in income inequality will hinder the abatement impact of renewable energy technological innovations on per capita emissions
Liu et al. (2020),	To answer the question of how income inequality influence energy efficiency	33 Belt and Road Initiative countries	The findings showed that energy efficiency within the selected countries was generally low over the entire study period.
Tan and Uprasen (2021)	A study on the asymmetric effect of income inequality on renewable energy consumption	Association of Southeast Asian Nations (ASEAN) for the period 1990 to 2015	The results showed that by reducing income inequalities, the demand and consumption of renewable energy increase in the long run.
Asongu and Odhiambo (2021)	Examine the linkages between financial development, renewable energy consumption, and income inequality	39 Sub-Saharan African countries	The results show that whereas financial development encourages renewable energy consumption, income inequality hinders the underlying positive impacts.
Panel (B) Corruption, Income Inequality and (non)renewable energy consumption			
Asongu et al. (2019) and	Assessing the conditional relationship between renewable energy and environmental quality in sub-Saharan Africa.	40 African countries for the period 2002 to 2017; fixed effects regressions and quantile fixed effects regressions	The findings show that renewable energy consistently decreases carbon dioxide (CO2) emissions. The negative effect is a decreasing function of CO2 emissions or the negative effect of renewable energy on CO2 emissions decreases with increasing levels of CO2 emissions.
Lisciandra & Migliardo (2017)	An empirical study of the impact of corruption on environmental performance: Evidence from panel data	A panel dataset of 153 countries from 2002 to 2012.	Corruption promotes environmental degradation by reducing the positive effect of renewable energy on the environment's quality and increasing the negative repercussions of nonrenewable energy consumption.
Lopez and Mitra (2000)	Corruption, pollution, and the Kuznets environment curve.	Theoretical studies	Corruption is not likely to preclude the existence of an inverted-U-shaped-Kuznets environmental curve under both these cases. For any level of per capita income the pollution levels corresponding to corrupt behaviour are always above the socially optimal level.
Boyce (1994)	To examine inequality as a cause of environmental degradation	Theoretical paper	Greater inequalities of power and wealth lead to more environmental degradation
Policardo et al. (2019).	Causality between income inequality and corruption in OECD countries	34 OECD countries between period 1995 and 2011	Corruption increases income inequality and income inequality affects corruption positively.
Chong and Calderon (2000)	Examines institutional quality and poverty measures in a cross-section of countries	least squares and two-stages least squares estimate over the period 1960 to 1990	Well-established institutions lower the degree, severity and incidence of poverty and income inequality.
Opeyemi et al. (2019)	Renewable energy, trade performance and the conditional role of finance and institutional capacity in sub-Sahara African countries	42 SSA countries for the period 2004–2016, and system GMM technique	Finance and regulation improves renewable energy and total trade nexus Corruption control matters is critical to improving the effect of renewable energy use on trade outcomes.

that the public sector redistributes income in developing countries to ensure acceptable equity between people. By contrast, there is evidence of high levels of corruption in the public sector within these developing countries (Gustavson and Sundström, 2018), which widens public sector debt levels (Benfratello et al., 2018) and eventually burdens the poor.

Corruption increases costs associated with the adoption of renewable energies through indirect payments to gratify the selfish interests of public officials and other illegal fiscal charges (Njoh et al., 2019), which discourages entrepreneurial engagements. That is, corruption increases the cost of doing business through informal payments and other illegal payments meant to facilitate the importation, registration, and approval of technologies in poorly regulated, corrupt countries. These costs lead to productivity and firm competitiveness reductions usually associated with adopting renewable energies (Opeyemi et al., 2019). Another way corruption shifts the consumption of renewable energies towards nonrenewable is through stalling of well-laid-out renewable energy policies, adversely affecting economies' economic welfare (Gennaioli and Tavoni, 2016). Partnership between the public and private sector industries is ineffectively channelled and used in countries with weak control over corruption (Opeyemi et al., 2019). The aggregate impact is stagnation of economic growth and a reduction in productivity.

Where there is greater corruption amongst policymakers, it reduces the stringency of energy policies. An increase in corruption moves the government's relative weighting away from welfare towards bribes, propelling the purchase of influence by making it affordable (Fredriksson et al., 2004). The more corruption persists, the stronger it impedes energy efficiency amongst countries in the Organization for Economic Co-operation and Development (Fredriksson et al., 2004).

Income inequality may influence the consumption of renewable energies by affecting social norms such as individualism, short-term interests, and consumerism (Uzar, 2020). Corruption represents greed and the desire of a person or group of persons to take illicit payments to the detriment of the common interest of society. This is the characteristic of societies with widening income inequality, as there is a general lack of social cohesion and collective action. In economies with corruption, there is widening income inequality, and such societies have low sensitivity to environmental issues (Boyce, 1994), exhibit rent-seeking attitudes, and typically use nonrenewable energy resources to gratify their interests (Berthe and Elie, 2015). There is a general lack of concern about the future repercussions of their current consumption decisions. These societies are typically identified by clear consumption and competition for status, where the motivation for consumption is to sustain a certain level of social status (Berthe and Elie, 2015). It is not surprising to observe that corruption is typical in societies with wide income inequalities (Policardo et al., 2019). Leaders seek to manipulate the system for their gains to meet their status in society. In the instance where consumption behaviours do not promote the use of clean energy, they are inclined to put a lot of pressure on the environment (Berthe and Elie, 2015)

Income inequality could affect renewable energy consumption through its impact on the quality of institutions and the power association between various societal groups. According to Sequeira and Santos (2018), institutions are significant in the determination of energy consumption, distribution of income, and environmental policies. Where there is income inequality, it could represent weak institutions (Chong and Calderon, 2000), and these weak institutions are a disincentive to effective investments (Mehrara et al., 2015) and could hamper the effectiveness of environmental policies. Within an environment where institutions are weak, high-class groups could engage in operations that deter a favourable environment (Uzar, 2020). The low levels of investments in renewable energy also affect the behaviour of people to use clean energy. There is the evolvement of power inequalities in determining renewable energy consumption and the way institutions work. There is also a rent-seeking attitude to optimize profit, making it harder for poorer individuals to enjoy renewable energy consumption (Strunz et al., 2016). In Table 1, Panel B, we summarise the empirical studies on the effect of corruption on primary and (non)renewable energy consumption. From the theoretical and empirical literature, we test the following hypothesis:

3. Methodology and data

3.1. Empirical models and estimation strategy

This study contributes to energy transition and net-zero emissions literature by investigating the impact of corruption, income inequality, and redistribution on renewable, nonrenewable, and total energy consumption. Following previous studies such as Uzar (2020) and Awaworyi Churchill et al. (2021), we specify a general dynamic model for estimating the impact of corruption, income inequality, and redistribution on renewable, nonrenewable, and total energy consumption.

$$lnEC_{it} = \alpha_0 lnEC_{it-1} + \beta_1 lnRGDPC_{it} + \beta_2 lnGINI_{it} + \beta_3 lnREDIST_{it} + \beta_4 CORRUP_{it} + \varnothing_1 X_{it} + \varepsilon_{it} \quad (1)$$

From Eq. (1), $lnEC_{it}$, is an indicator of the energy variables (renewable, nonrenewable, and total energy consumption) of country i at time t. $lnRGDPC_{it}$, is an indicator of GDP per capita country i at time t. $lnGINI_{it}$, is an indicator of income inequality of country i at time t. $lnREDIST_{it}$, is an indicator of income redistribution of country i at time t. $lnCORRUP_{it}$, is an indicator of control of corruption of country i at time t. lnX_{it} is an indicator of control covariates (trade openness, foreign direct investment, education, inflation, and foreign aid) that affect energy consumption in country i at time t. α_0 is the unknown coefficient of the lagged energy variables, $\beta_1 - \beta_4$ are the unknown coefficients of the explanatory variables, $\varnothing_1$ is the unknown coefficient of the control covariates and ε_{it} is the error term.

To further contribute to the political economy literature, we examine if corruption conditions income inequality and redistribution to either increase or decrease the consumption of the energy variables (renewable, nonrenewable, and total energy consumption). Theoretically, corruption is argued to distort income inequality and redistribution by inducing tax evasion and exemption (Gupta et al., 2002; Keneck-Massil et al., 2021). Tax evasion and exemption enjoyed by the wealthy make the tax burden fall on the poorer people (Keneck-Massil et al., 2021). Also, corruption can divert resources allocated for implementing pro-poor policies and programs towards policies that may enhance the income of the wealthy at the expense of the poor (Glaeser et al., 2003; Keneck-Massil et al., 2021). Several empirical studies have revealed that corruption worsens income inequality or impedes income redistribution (Glaeser et al., 2003; Keneck-Massil et al., 2021; Li et al., 2000). We, therefore, hypothesise that corruption conditions income inequality and redistribution to affect energy consumption. With regards, Eq. (1) is augmented with the interaction terms of corruption, income inequality, and redistribution. Eq. (2) is, therefore, used to examine the interactive effect of corruption and income inequality on the energy variables, while Eq. (3) is used to model the interactive impact of corruption and income redistribution on the energy variables.

$$lnEC_{it} = \alpha_0 lnEC_{it-1} + \beta_1 lnRGDPC_{it} + \beta_2 lnGINI_{it} + \beta_3 lnREDIST_{it} + \beta_4 CORRUP_{it} + \delta_1 (CORRUP \times lnGINI)_{it} + \varnothing_1 X_{it} + \varepsilon_{it} \quad (2)$$

$$lnEC_{it} = \alpha_0 lnEC_{it-1} + \beta_1 lnRGDPC_{it} + \beta_2 lnGINI_{it} + \beta_3 lnREDIST_{it} + \beta_4 CORRUP_{it} + \delta_2 (CORRUP \times lnREDIST)_{it} + \varnothing_1 X_{it} + \varepsilon_{it} \quad (3)$$

Regarding the estimation strategy, this study utilised the dynamic econometric technique to estimate the dynamic models specified above. Thus, the Blundell and Bond (1998) two-step dynamic system-generalised method of the moment (GMM) was deployed as the primary estimator in this study. In dynamic models, it is demonstrated that the lag of the independent variable correlates with the unobserved effects and that estimating such models with traditional estimation techniques, such as ordinary least squares, could render the estimates inefficient (Arellano and Bond, 1991). It is recommended that differencing dynamic models to remove unobserved country and time effects would lead to the presentation of efficient estimates. To present unbiased estimates from dynamic models, Arellano and Bond (1991) developed the first difference generalised method of moment technique for removing unobserved country and time effects using the first differencing transformation. Nevertheless, it has been shown that when time series observation is small, and the autoregressive parameter is relatively large, the Arellano and Bond (1991) first difference generalised method of moment technique yields inefficient estimates because of poor precision and having large finite sample bias (see, Blundell and Bond, 1998, p. 115). In view of that, the dynamic system-GMM technique was developed by Blundell and Bond (1998) to address the limitations in the first difference generalised method of moment technique. When differencing, the Blundell and Bond (1998) system generalised method of moment technique uses the lagged differences of the dependent variable as instruments for equations in levels. It includes the lagged levels of the dependent variable as instruments for equations in first differences. Since this study involves a large number of countries ($N = 166$) and a relatively shorter time ($T = 28$), we find the Blundell and Bond (1998) two-step dynamic system-GMM a more suitable estimator since it has proven to provide reliable estimates with smaller T and large N. We evaluate the reliability of the Blundell and Bond (1998) two-step dynamic system-generalised method of moment estimates using the Hansen test, first (AR(1)) and second-order (AR(2)) autocorrelation.

3.3. Data

Based on data availability, this study examines the impact of corruption, income inequality, and redistribution on renewable and nonrenewable and total energy consumption using a comprehensive panel dataset for 166 countries (see Appendix Table A1) between 1990 and 2017.[1] The data used in this study are drawn from multiple sources such as the World Development Indicators (WDI), Standardized World Income Inequality Database (SWIID), and World Governance Indicators (WGI). The variables deployed in this study as described as follows:

- ***Dependent variables***: Three main energy variables were used as dependent variables. The first energy variable, renewable energy consumption, was measured using renewable energy consumption (% of total final energy consumption). Renewable energy consumption is the share of renewable energy in total final energy consumption. The second energy variable, nonrenewable energy consumption, was represented using fossil fuel energy consumption (% of total). Nonrenewable energy comprises coal, oil, petroleum, and natural gas products. Finally, total energy consumption was represented using energy use (kg of oil equivalent per capita). Total energy consumption refers to the use of primary energy before transformation to other end-use fuels, equal to indigenous production plus imports and stock changes, minus exports and fuels supplied to ships and aircraft engaged in international transport. The reason behind using three different energy variables as dependent variables is first motivated by the argument that low-income and more affluent households spend differently on renewable and nonrenewable energy. Wealthy households and economies could easily transition to a renewable energy economy as they have the financial resources to bear the initial cost of renewable energy investment compared to poorer households or economies. This indicates that poorer households spend more on nonrenewable energy than renewable energy. Therefore, using renewable and nonrenewable energy separately for the analysis would help understand the validity of this argument and appropriately inform the design. The second reason is that using only total energy consumption as a dependent variable would obscure policymakers' understanding of how corruption, income inequality and redistribution affect different sources of energy consumption—renewable and nonrenewable—and this could lead to the design and implementation of inefficient energy policies in the face of poor-socioeconomic conditions. These energy variables were drawn from the WDI database.
- ***Independent variables:*** Corruption, income inequality, and redistribution are the key variables of interest. Corruption was represented with control of corruption from the World Governance Indicators (WGI). The control of corruption index captures perceptions of the extent to which public power is exercised for private gain, including petty and grand forms of corruption and "capture" of the state by elites and private interests. The corruption control index ranges from −2.5 to 2.5 (Kaufmann et al., 2011). The higher the score, the less corruption activities, and vice versa. The income inequality and redistribution data were drawn from Solt's (2016) SWIID to represent income inequality. The income inequality variable was represented with net (post-tax/post-transfer Gini) income inequality and market (pre-tax/pre-transfer Gini) income inequality. Also, income redistribution was measured using absolute and relative variables. Absolute income redistribution is the number of Gini-index points where market-income inequality is reduced due to taxes and transfers and is the difference between market and net income inequality (Solt, 2016). Relative income redistribution is the percentage reduction in market-income inequality due to taxes and transfers and is calculated as the difference between market-income inequality and net income inequality divided by market-income inequality multiplied by 100 (Solt, 2016). Theoretically, the intuition behind the measurements of these income inequality and distribution variables are different and have different implications for policy. Therefore, comparatively analysing the effect of the net (post-tax/post-transfer Gini) and market (pre-tax/pre-transfer Gini) versions of income inequality as well as relative and absolute versions of income redistribution, would yield important insight into the relationship between income inequality, redistribution and the energy mix as well as policy design. This study relied on the SWIID database because it combines and standardises inequality data from well-established income inequality databases such as the World Income Inequality database, Luxembourg Income Studies, and World Income Distribution Data, amongst others (see Solt, 2016).
- ***Control variables:*** Consistent with the literature on the determinants of energy consumption (including renewable and nonrenewable energy consumption), we included variables such as economic growth, trade openness, foreign direct investment, education, inflation, and foreign aid in the energy models (Anton and Afloarei Nucu, 2020; Awaworyi Churchill et al., 2021; Dogan et al., 2022; Sadorsky, 2009, 2011b; Soytas and Sari, 2003, 2009; Uzar, 2020). Economic growth is measured using GDP per capita (constant 2015 US$); trade openness is measured with trade (% of GDP); inflation is represented with GDP deflator (annual%); education is represented with secondary school enrolment (% gross), foreign direct investment is measured with foreign direct investment, net inflows (% of GDP) and foreign aid is represented with net official development assistance and official aid received (constant 2020 US$). All the control variables were drawn from WDI.

Table 2 shows the descriptive statistics for the variables used for the analysis. Except for the control of the corruption variable, the remaining variables are transformed using natural logarithms; hence, the descriptive statistics for these variables are natural logarithms values. The mean values for total, nonrenewable, nonrenewable, and renewable energy consumption are 7.114, 4.035, and 2.827, respectively. These statistics show that the share of nonrenewable energy in the global total energy mix exceeds that of renewable energy in total global energy. Regarding income inequality and redistribution variables, the post-tax/transfer Gini Index and pre-tax/transfer Gini Index have a mean of 3.633 and

Table 2
Descriptive statistics.

Variable	Definitions	Mean	Std. Dev.	Min	Max
Lnener	Total energy consumption	7.114	1.035	2.256	10.004
Lnfosil	Nonrenewable energy consumption	4.035	0.707	0.495	4.605
Lnrenew	Renewable energy consumption	2.827	1.592	−6.340	4.588
lngini_disp	Post-tax/transfer Gini Index	3.633	0.227	2.896	4.148
lngini_mkt	Pre-tax/transfer Gini Index	3.832	0.139	3.332	4.282
lnrel_red	Relative redistribution	2.756	1.033	−1.609	3.980
lnabs_red	Absolute redistribution	1.995	1.031	−2.303	3.254
Corrup	Control of corruption	−0.059	0.994	−1.723	2.470
Lnrgdpc	Real GDP per capita	8.272	1.471	5.087	11.626
Lnfdi	Foreign direct investment	0.793	1.557	−12.507	6.113
Lntrade	Trade openness	4.303	0.523	2.406	6.093
Lninfla	Inflation	1.762	1.423	−6.701	10.195
Lnoda	Foreign aid	19.169	1.585	9.210	23.121
Lnsse	Education	4.177	0.623	1.651	5.099

[1] As of the time of writing this paper, Standardized World Income Inequality Database (SWIID), the main sources for the income inequality and redistribution data, which are the key explanatory variables of interest, were only up to 2017.

3.832, respectively, and the mean for relative income redistribution and absolute income redistribution are 2.756 and 1.995, respectively. These values show that income inequality before tax is relatively higher than income inequality after tax, but the differences are marginal. This indicates that income redistribution through tax and transfers has contributed to the marginal decline of global income inequality. Control of corruption has a mean value of −0.059, indicating that, on average, corruption activities based on citizens' perceptions are high for the representative sample. The mean for real GDP per capita, foreign direct investment, trade openness, inflation, foreign aid, and education are 8.272, 0.793, 4.303, 1.762, 19.169, and 4.177, respectively. The multicollinearity matrix is presented in Appendix Table A2; it is obvious that there is no issue of multi-collinearity.

4. Empirical analysis and discussion

4.1. The impact of corruption, income inequality, and redistribution on nonrenewable and renewable energy consumption

This section presents the empirical discussion of our results. While our focus is to analyse the impact of corruption, income inequality, and redistribution on renewable and nonrenewable energy consumption, we first undertake a benchmark estimation focusing on total energy consumption. Thus, Models 1 to 4 of Table 3 display the results of total energy consumption, whereas Models 5 to 8 and 9 to 12 show the findings for nonrenewable and renewable energy consumption, respectively. The findings show that the post-tax/transfer Gini Index (*lngini_disp*) exerts a negative and statistically significant effect on total energy consumption (see Models 1). Income inequality determines capacity; therefore, since lower-income groups spend a larger proportion of their income on energy, any factor that increases the cost of energy production, such as pollution tax, falls on the lower-income group (see Kaasa, 2003). The relative tax burden would be heavier on the poor or individuals with fixed nominal income. Unsurprisingly, post-tax income inequality reduces overall energy consumption (see Sari and Soytas, 2007). We find a similar outcome for pre-tax/transfer Gini Index (*lngini_mkt*)—as shown in Model 2. Overall, our findings align with prior literature arguing that taxes widen income inequality (see Johansson and Gustafsson, 1997), which reduces real income—through its effect on real disposal income—and overall energy consumption (see Masih and Masih, 1996). However, when looking at the impact separately, the market income inequality has a much higher reducing effect on total energy consumption than the post-tax income inequality.

To understand the findings in Models 1 to 4, we disaggregated total energy consumption into renewable and nonrenewable consumption. First, in models 5 to 8, we find that the post-tax/transfer Gini index (*lngini_disp)* has a significant positive impact on nonrenewable energy consumption similar to the estimated coefficient for pre-tax/transfer Gini index (*lngini_mkt)*. This result is unsurprising since most in developing countries with profound income inequality consume cheap energy, usually unclean or environmentally unfriendly (see Dogan and Seker, 2016; Güney, 2019). When examined from a renewable energy consumption perspective, we find that the post-tax/transfer Gini Index (*lngini_disp*) and pre-tax/transfer Gini Index (*lngini_mkt*) reduce renewable energy consumption. This reinforces our earlier argument that poverty and increasing income inequality reduce the demand for clean energy, especially in developing countries where renewables are expensive (Thiam, 2011). These findings generally suggest that poor households prefer nonrenewable because it is less expensive in terms of installation and very easy to access (see Martinot et al., 2002).

We also examined the role of income redistribution on energy consumption and found that relative income redistribution (*lnrel_red*) and absolute income redistribution (lnabs_red*)* enter the estimation with a positive coefficient in the model with total energy consumption. Unlike the Gini indices, relative income redistribution (*lnrel_red*) and absolute income redistribution (*lnabs_red)* are found to reduce nonrenewable energy consumption. The reduction in market income inequality by personal income taxes serves as an incentive for consumers to change their energy preferences (see Uzar, 2020). The estimates indicate that relative income redistribution (*lnrel_red*) and absolute income redistribution (lnabs_red*)* increase renewable energy consumption, but the impact is statistically significant for absolute income redistribution. This finding implication aligns with the assumption that income redistribution could promote renewable energy consumption and facilitate energy transition.

Central to this study is the role of corruption in energy consumption. The estimates show that control of corruption has a significant negative relationship with total energy consumption (see Models 1–4). The estimates show that when both post-tax/transfer Gini and pre-tax/transfer Gini Indices are controlled, control of corruption has a negative relationship with nonrenewable energy consumption (see Models 5–6); however, when income redistribution variables are in the model, control of corruption exert a significant positive relationship with nonrenewable energy consumption (see Models 7–8). The implication is that when corruption widens income inequality, poorer households tend to be energy-efficient, and thus, leads to a reduction in total energy consumption. However, when control of corruption facilitates the redistribution of income, it induces households to increase total energy usage because of the increase in income. As presented in Models 9–12, control of corruption has a significant positive relationship with renewable energy consumption, indicating that addressing corruption would facilitate energy transition. Generally, results show that control of corruption improves energy efficiency and increases renewable energy by encouraging long-term investment in renewable energy projects. This result agrees with Uzar's (2020) and Sequeira and Santos's (2018) empirical findings that control of corruption drives renewable energy consumption.

Concerning the control variables, the findings reveal a positive effect of real GDP per capita (*lnrgdpc*) on total energy consumption (see Models 1 & 2) and nonrenewable energy consumption (see Models 5–8). This result implies that increasing economic growth drives total and nonrenewable energy consumption, which is consistent with the findings of Chiou-Wei et al. (2008). Contrarily, we observed that the impact of GDP per capita is influenced by income inequality and redistribution. For instance, in models with income inequality (Models 9 and 10), real GDP per capita has a negative and significant effect on renewable energy consumption. These findings suggest that with high-income inequality, GDP per capita reduces renewable energy consumption since income inequality impedes economic growth (Alesina and Rodrik, 1994; Barro, 2000). Therefore, poor economic growth implies lower household income, thereby reducing renewable energy consumption. Also, in models with income redistribution variables (Models 11–12), GDP per capita has a significant positive effect on renewable energy consumption, suggesting that with income redistribution policies, demand for renewable energy increases due to improvement in poorer households' income increases. The positive effect of GDP per capita on total energy and nonrenewable energy consumption has also been documented in several studies (Murshed and Alam, 2021; Sadorsky, 2010; Zhao et al., 2020).

We also find that in the models with income inequality variables, foreign direct investment (*lnfdi)* significantly increases total energy consumption (see Models 1 & 2), while in models with income redistribution, foreign direct investment significantly reduces total energy consumption. This result signifies that in the presence of high-income inequality, foreign direct investment is associated with increasing energy consumption intensity; however, with income distribution, foreign direct investment improves energy efficiency by reducing energy consumption intensity. Thus, income inequality could impede foreign direct investment, limiting the role of foreign direct investment in reducing energy consumption intensity. Also, foreign direct investment (*lnfdi)* exerts a significantly negative effect on nonrenewable energy consumption (see Models 5 & 6). In addition, foreign direct investment

Table 3
Effect of income inequality, redistribution and corruption on energy nonrenewable and renewable energy consumption.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
l.lnener	0.804***	0.863***	0.654***	0.653***								
	(0.003)	(0.003)	(0.042)	(0.049)								
l.lnfosil					0.779***	0.770***	0.984***	1.013***				
					(0.002)	(0.001)	(0.030)	(0.032)				
l.lnrenew									0.809***	0.804***	0.931***	0.923***
									(0.003)	(0.002)	(0.026)	(0.026)
lnrgdpc	0.073***	0.027***	0.071	0.060	0.042***	0.052***	0.029**	0.040***	−0.059***	−0.075***	0.115***	0.109**
	(0.004)	(0.002)	(0.052)	(0.051)	(0.002)	(0.003)	(0.012)	(0.013)	(0.005)	(0.006)	(0.043)	(0.045)
lnfdi	0.003***	0.003***	−0.024***	−0.028***	−0.004***	−0.004***	0.004	0.004	0.010***	0.012***	0.009	0.008
	(0.000)	(0.000)	(0.008)	(0.008)	(0.000)	(0.000)	(0.003)	(0.003)	(0.001)	(0.001)	(0.009)	(0.008)
lntrade	0.014***	0.002	0.175***	0.186***	0.034***	0.030***	−0.010	−0.009	−0.083***	−0.101***	−0.028	−0.038
	(0.003)	(0.003)	(0.029)	(0.032)	(0.002)	(0.001)	(0.008)	(0.009)	(0.005)	(0.005)	(0.023)	(0.024)
lninfla	0.016***	0.013***	0.028***	0.036***	0.003***	0.001***	0.006***	0.006***	−0.013***	−0.013***	−0.007	−0.009*
	(0.000)	(0.000)	(0.007)	(0.007)	(0.000)	(0.000)	(0.001)	(0.002)	(0.001)	(0.001)	(0.005)	(0.005)
lnoda	−0.000	0.001***	−0.006	−0.004	0.015***	0.011***	0.007***	0.008***	0.008***	0.007***	0.016	0.010
	(0.000)	(0.000)	(0.014)	(0.014)	(0.001)	(0.000)	(0.003)	(0.003)	(0.001)	(0.001)	(0.010)	(0.009)
lnsse	0.097***	0.093***	0.332***	0.403***	0.136***	0.136***	−0.076*	−0.129***	−0.167***	−0.157***	−0.385***	−0.444***
	(0.005)	(0.004)	(0.098)	(0.104)	(0.002)	(0.003)	(0.042)	(0.046)	(0.008)	(0.010)	(0.135)	(0.150)
corrup	−0.006**	0.042***	−0.072***	−0.038*	−0.000	−0.045***	0.039***	0.044***	0.088***	0.115***	0.052	0.044
	(0.003)	(0.002)	(0.023)	(0.021)	(0.002)	(0.001)	(0.013)	(0.013)	(0.005)	(0.004)	(0.059)	(0.055)
lngini_disp	−0.201***				0.153***				−0.001			
	(0.014)				(0.013)				(0.019)			
lngini_mkt		−0.241***				0.215***				−0.223***		
		(0.053)				(0.016)				(0.026)		
lnrel_red			0.069***				−0.013**				0.015	
			(0.025)				(0.006)				(0.016)	
lnabs_red				0.032				−0.010*				0.040***
				(0.030)				(0.006)				(0.015)
Constant	0.999***	1.209***	−0.380	−0.556	−1.008***	−1.225***	0.085	0.055	2.003***	3.043***	0.689	1.161**
	(0.068)	(0.225)	(0.531)	(0.606)	(0.066)	(0.063)	(0.114)	(0.132)	(0.114)	(0.126)	(0.477)	(0.516)
Observations	851	851	398	398	812	812	398	398	1058	1058	421	421
Hansen	90.730	89.306	21.669	21.057	75.596	76.448	18.125	19.674	100.473	98.882	22.236	21.107
P(Hansen)	0.518	0.560	0.359	0.394	0.892	0.879	0.579	0.478	0.412	0.456	0.328	0.391
AR (1)	0.000	0.000	0.001	0.001	0.008	0.008	0.008	0.008	0.015	0.015	0.037	0.039
AR (2)	0.557	0.578	0.208	0.324	0.851	0.806	0.852	0.798	0.507	0.502	0.915	0.913
No. countries	166	166	166	166	166	166	166	166	166	166	166	166
No. of instruments	102	102	30	30	102	102	30	30	108	108	30	30

Standard errors in parentheses. Hansen-test refers to the over-identification test for the restrictions in system-GMM estimation. The AR (1) and AR (2) tests are the Arellano–Bond tests for first and second-order autocorrelation in first differences.

* $p < 0.10$,.

** $p < 0.05$,.

*** $p < 0.01$.

(*lnfdi)* significantly improves renewable energy consumption (see Models 9 & 10). These findings suggest that foreign direct investment is a panacea for transitioning towards renewable energy while reducing nonrenewable energy consumption. This finding aligns with the argument that foreign investment has been the main source of funding renewable energy projects across the globe (Paramati et al., 2017). Also, foreign investment is associated with reducing nonrenewable energy consumption since it enhances the diffusion of advanced and clean energy technologies across countries. Some studies, including Paramati et al. (2017) and Paramati et al. (2016), have shown that foreign direct investment is positively and significantly associated with renewable energy consumption. Our result is similar to the Doytch and Narayan (2016) results, proving that foreign direct investment reduces nonrenewable energy consumption while increasing renewable energy consumption.

Trade openness (*lntrade)* is observed to increase both total energy consumption substantially (see Models 1, 3 & 4) as well as nonrenewable energy consumption (see Models 5 & 6). On the other hand, trade openness (*lntrade)* has a statistically significant negative effect on renewable energy (see Models 9 & 10). The positive effect of trade openness on total energy and nonrenewable energy consumption shows that scale effect, suggesting that trade openness increases total energy and nonrenewable energy demand as it contributes to increasing production (Cole, 2006). On the other hand, the negative relationship between trade openness and renewable energy consumption indicates that trade has not been associated with the transfer of technologies that stimulate the growth of renewable energy demand. The results of the positive effect of the relationship between trade openness and total energy and nonrenewable energy consumption align with the findings of Kyophilavong et al. (2015), Sadorsky (2011b) and Sadorsky (2012) but contradict the findings of Sbia et al. (2014). Also, the negative relationship between trade openness and renewable energy consumption diverges from the findings of Zhang et al. (2021) and Omri and Nguyen (2014).

Similarly, we find inflation (*lninfla)* to increase total energy consumption significantly (see Models 1–4) as well as nonrenewable energy consumption (see Models 5–8). However, inflation (*lninfla)* negatively relates to renewable energy consumption (see Models 9,10 & 12). This result explains the substitutability between renewable and nonrenewable energy. Comparatively, nonrenewable energy is relatively cheaper than renewable energy. Also, given the close substitutability between renewable and nonrenewable energy, when the price level in the economy increases, people would demand more nonrenewable energy and less renewable energy. Previous studies, such as Sadorsky (2011a), show that inflation is negatively related to total energy consumption, contradicting our results on the positive linkage between inflation and total energy and nonrenewable energy consumption.

The estimates also show that foreign aid (*lnoda)* has a significant positive impact on total energy consumption (see Model 2), nonrenewable energy consumption (see Models 5–8) as well as renewable energy consumption (see Models 9 & 10). This finding implies that although foreign aid increases nonrenewable energy, it increases renewable energy consumption and energy efficiency. The positive effect of foreign aid on renewable energy has also been documented in the study of Villanthenkodath et al. (2023). Contrarily to the finding of Yao et al. (2019), our results suggest that education (*lnsse), as a proxy of human capital,* is observed to have a statistically significant positive effect on total energy consumption (see Models 1–4). At the same time, the results suggest that in the models with income inequality variables, education (*lnsse)* is observed to increase nonrenewable energy consumption significantly (see Models 5–6). In contrast, in models with income redistribution (see Models 7–8), education (*lnsse)* significantly reduces nonrenewable energy consumption. Also, education (*lnsse)* is observed to have a statistically significant negative effect on renewable energy consumption (see Models 9–12). These results highlight that income inequality and redistribution influence the relationship between human capital and (non)renewable energy consumption. Overall, education (human capital) does not favour renewable energy consumption and energy efficiency, as the literature argues (see Yao et al., 2019).

4.2. Accounting for regional dummy variables

Across the globe, a significant difference exists in institutional quality (corruption), income inequality and energy consumption amongst geographical regions. We, therefore, extend the analysis by including regional dummy variables in our regression. The regional dummy variables enable us to control regional heterogeneity in our sample. We, therefore, follow the World Bank geographical regional classification to group the countries included in this study: Sub-Saharan Africa (SSA), East Asia & Pacific (EAP), Middle East and North Africa (MENA), Europe and Central Asia (ECA), and South Asia (SAR) and Latin America & Caribbean (LAC).[2] Table 4 presents the estimated results after accounting for regional dummy variables. Latin America & Caribbean (LAC) was used as the base/control region for comparison purposes. The result for the regional dummy variables suggests that SSA economies significantly contribute less to total energy consumption (see Models 1 & 2) than the control group (LAC). With regard to nonrenewable energy consumption, SSA is not statistically different from LAC. Also, SSA economies have higher renewable energy consumption (see Models 9 & 10) than the base region (LAC). These findings show that in SSA, renewable energy consumption is significantly higher. At the same time, the intensity of total energy consumption is significantly less. These findings can be attributed to the vast endowment of renewable energy resources, including critical minerals in SSA. For instance, the IEA (2022) report shows that SSA has over 40% of global reserves of critical minerals (cobalt, manganese, and platinum). Our result further supports the observation that SSA has the lowest average primary (total) energy consumption globally, reflecting less primary energy use in production compared to other regions such as LAC. The EAP dummy has a significant negative effect on total energy consumption (Model 1). Also, the EAP dummy contributes significantly to nonrenewable energy consumption [Models 5–6], while it only has a statistically significant negative effect on renewable energy consumption in Model 12. This shows that EAP's primary and renewable energy consumption is relatively less than primary and renewable energy consumption in LAC. At the same time, nonrenewable energy consumption is relatively higher than LAC. The relatively higher consumption of nonrenewable energy in EAP can be attributed to their rapid economic growth, over-reliance on coal, higher population size, industrialization, and limited domestic energy resources in some countries.

Regarding the MENA region, our findings show that the MENA region dummy variable has a significant positive effect on total energy consumption (see Models 1–4) but has a statistically significant negative effect on renewable energy consumption (see Models 9–12). Broadly, MENA has an insignificant effect on nonrenewable energy consumption (see Models 5–8). These show that total energy consumption has been significantly higher in MENA than in LAC. At the same time, renewable energy consumption is less in MENA than in LAC. This clearly shows that the MENA region relies significantly on coal, oil, petroleum, and natural gas products because it is a major nonrenewable energy producer. In addition, higher economic output in the MENA region may have contributed to the significant increase in total energy consumption. The results also show that the ECA dummy variable has a significant positive effect on total energy consumption (see Models 1–4) than LAC. On the other hand, the ECA dummy variable has a statistically significant negative effect on nonrenewable energy consumption (see Models 5–6). In addition, the ECA dummy positively affects renewable energy

[2] We selected Latin America & Caribbean as the base region because it is the developing region with high income inequality (Busso, 2020) and also has relatively higher energy consumption intensity.

Table 4
Effect of income inequality, redistribution and corruption on energy nonrenewable and renewable energy consumption, including regional dummies.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
L.lnener	0.714***	0.758***	0.426***	0.444***								
	(0.007)	(0.006)	(0.071)	(0.070)								
L.lnfosil					0.714***	0.716***	0.924***	0.952***				
					(0.003)	(0.002)	(0.063)	(0.072)				
L.lnrenew									0.653***	0.691***	0.763***	0.759***
									(0.004)	(0.004)	(0.067)	(0.066)
lnrgdpc	0.166***	0.158***	0.238**	0.241**	0.023***	0.002	−0.017	0.000	0.014	0.043***	0.233***	0.246***
	(0.008)	(0.007)	(0.102)	(0.099)	(0.006)	(0.005)	(0.046)	(0.048)	(0.014)	(0.015)	(0.081)	(0.081)
lnfdi	−0.003***	−0.006***	0.020**	0.024***	−0.000	−0.003***	0.008	0.010	0.013***	0.013***	0.003	0.014
	(0.001)	(0.001)	(0.009)	(0.009)	(0.001)	(0.001)	(0.007)	(0.007)	(0.002)	(0.002)	(0.023)	(0.025)
lntrade	−0.000	0.002	0.115**	0.127**	0.020***	0.033***	0.002	−0.005	−0.096***	−0.072***	−0.125*	−0.188**
	(0.004)	(0.002)	(0.050)	(0.050)	(0.004)	(0.007)	(0.035)	(0.036)	(0.017)	(0.014)	(0.076)	(0.080)
lninfla	0.024***	0.021***	0.004	0.002	0.008***	0.008***	−0.004	−0.006	−0.035***	−0.031***	−0.014	−0.025
	(0.000)	(0.000)	(0.008)	(0.008)	(0.000)	(0.000)	(0.006)	(0.006)	(0.001)	(0.001)	(0.019)	(0.017)
lnoda	−0.002***	0.005***	−0.018	−0.020	0.012***	0.014***	0.012	0.011	0.041***	0.033***	−0.033	−0.049*
	(0.000)	(0.000)	(0.014)	(0.015)	(0.001)	(0.001)	(0.009)	(0.010)	(0.002)	(0.002)	(0.029)	(0.030)
lnsse	−0.126***	−0.071***	0.122	0.129	0.241***	0.265***	−0.126	−0.162	−0.200***	−0.171***	0.072	0.018
	(0.005)	(0.007)	(0.216)	(0.203)	(0.003)	(0.004)	(0.135)	(0.140)	(0.020)	(0.016)	(0.319)	(0.316)
corrup	−0.012***	−0.019***	−0.254***	−0.260***	0.020***	0.023***	0.006	0.003	0.040***	0.004	−0.088	−0.146
	(0.003)	(0.002)	(0.038)	(0.036)	(0.004)	(0.001)	(0.020)	(0.031)	(0.008)	(0.008)	(0.104)	(0.108)
SSA	−0.169***	−0.075*	0.291	0.366	0.003	−0.033	−0.204	−0.130	0.361***	0.404***	−0.098	0.089
	(0.040)	(0.039)	(0.394)	(0.403)	(0.023)	(0.027)	(0.395)	(0.388)	(0.043)	(0.046)	(0.808)	(0.887)
EAP	−0.272**	0.058	0.410	0.422	0.101***	0.091**	0.326	0.142	0.171	0.101	−2.443	−2.583*
	(0.118)	(0.068)	(0.460)	(0.450)	(0.028)	(0.045)	(0.647)	(0.517)	(0.136)	(0.091)	(1.796)	(1.477)
MENA	0.283***	0.152***	0.757**	0.659*	0.023	−0.078*	0.417	0.347	−0.681***	−0.524***	−2.292***	−2.272***
	(0.065)	(0.048)	(0.334)	(0.355)	(0.030)	(0.045)	(0.475)	(0.461)	(0.087)	(0.104)	(0.798)	(0.785)
ECA	0.209***	0.210***	0.342***	0.352***	−0.059***	−0.133***	−0.030	−0.002	0.006	0.045	0.241	0.317*
	(0.037)	(0.021)	(0.107)	(0.096)	(0.016)	(0.019)	(0.042)	(0.038)	(0.042)	(0.036)	(0.178)	(0.184)
SAR	0.023	−0.026	0.227	0.218	0.021	−0.013	−0.188	−0.123	0.220*	0.401***	1.222	1.160
	(0.073)	(0.043)	(1.125)	(1.223)	(0.026)	(0.033)	(0.516)	(0.509)	(0.120)	(0.101)	(2.381)	(2.263)
lngini_disp	−0.419***				0.179***				−0.001			
	(0.058)				(0.019)				(0.042)			
lngini_mkt		0.163***				0.139***				0.264***		
		(0.062)				(0.019)				(0.061)		
lnrel_red			0.082**				0.026				−0.192**	
			(0.036)				(0.029)				(0.078)	
lnabs_red				0.069**				0.020				−0.197**
				(0.034)				(0.029)				(0.080)
Constant	2.660***	−0.148	0.966	0.849	−1.008***	−0.882***	0.724*	0.670*	1.388***	−0.106	−0.043	0.494
	(0.253)	(0.289)	(0.754)	(0.713)	(0.124)	(0.108)	(0.386)	(0.400)	(0.296)	(0.403)	(0.934)	(0.987)
Observations	851	851	398	398	812	812	398	398	1058	1058	421	421
Hansen	81.715	83.925	17.687	18.418	77.324	75.926	10.126	10.794	95.079	97.571	19.316	18.320
P(Hansen)	0.640	0.573	0.543	0.495	0.762	0.796	0.812	0.767	0.421	0.353	0.200	0.246
AR (1)	0.000	0.000	0.018	0.018	0.007	0.007	0.015	0.017	0.013	0.013	0.025	0.020
AR (2)	0.413	0.459	0.122	0.129	0.655	0.661	0.784	0.769	0.353	0.367	0.864	0.736
No. of countries	166	166	166	166	166	166	166	166	166	166	166	166
No. of instruments	102	102	34	34	102	102	30	30	108	108	30	30

Standard errors in parentheses. Hansen-test refers to the over-identification test for the restrictions in system-GMM estimation. The AR (1) and AR (2) tests are the Arellano–Bond tests for first and second-order auto-correlation in first differences.

* $p < 0.10$,.

** $p < 0.05$,.

*** $p < 0.01$.

Note: For regional Comparison, Latin America and the Caribbean region is used as the baseline. SSA= sub-Saharan Africa, EAP = East Asia & Pacific, MENA = Middle East and North Africa, ECA =Europe and Central Asia and SAR =South Asia.

consumption but significantly only in Model 12. These results show that compared to LAC, primary energy consumption and renewable energy consumption are significantly higher in ECA than in LAC. However, nonrenewable energy consumption has been low in ECA compared to LAC. Increasing production and higher economic could explain ECA's higher primary energy consumption. Also, ECA's higher renewable and less nonrenewable energy consumption reflects their large-scale investment in renewable energy and commitment to supporting the global energy transition. The SAR dummy variable suggests that SAR economies are not statistically different from LAC in total and nonrenewable energy consumption. However, SAR has a statistically significant positive effect on renewable energy consumption. This result highlights that renewable energy consumption has been significantly higher in South Asia compared to Latin America and the Caribbean.

The results for the inequality measures did not change significantly even after accounting for the regional dummy variables, except for pre-tax/transfer Gini Index (*lngini_mkt*), which showed a positive effect on total energy consumption (see Model 2), nonrenewable energy consumption (see Model 6) and renewable energy consumption (see Model 10). The impact of control of corruption did not change as the negative effect on total energy consumption (see Table 4, Models 1–4) is estimated. Corruption distorts the gains from public expenditure on economic growth and worsens the operationalization of energy policies (see Wu et al., 2010). Notwithstanding, control of corruption exerts a statistically significant positive impact on nonrenewable energy consumption (see Models 5 & 6) and renewable energy consumption (see Model 9). We also observed that relative and absolute income redistribution exerts a statistically significant positive effect on total energy consumption (see Models 1–4) while significantly reducing renewable energy consumption (see Models 9–12). Both absolute and relative income redistribution significantly affect nonrenewable energy consumption. The changes in the magnitude of coefficients and signs on control of corruption, income inequality, and income redistribution might be due to variations in corruption, income inequality, and redistributive policies amongst geographical regions. Therefore, we extend our analysis in the next section, examining the interactive effect between the regional dummy variables and control of corruption, income inequality, and income redistribution on the energy variables.

Note that when regional dummies are controlled for in the estimation, the control variables did not change drastically in signs, although the magnitude of their coefficients changed in some cases. Specifically, the effect of real GDP per capita is consistent, whereas foreign direct investment (*lnfdi*) hurts total energy consumption (see Models 1 & 2) and nonrenewable energy consumption (see Models 6) only. Trade openness (*lntrade*) has a positive effect on both total energy consumption (see Models 3 & 4) and nonrenewable energy consumption (see Models 5 & 6). However, it exerts renewable energy is negative (see Model 9–10). The inflation (*lninfla*) coefficient is positive on total energy consumption (see Models 1 & 2) and nonrenewable energy consumption (see Models 5 & 6). Its impact on renewable energy consumption is detrimental (see Models 9 &10). The results for foreign aid (*lnoda*) and education (*lnsse*) remain consistent.

4.3. Do corruption, income inequality, and redistribution explain the variation in (non)renewable energy consumption between geographical regions?

To test if corruption, income inequality, and redistribution explain the variation in (non)renewable energy consumption between geographical regions, we examine the interaction effect between the regional dummy variables and corruption, income inequality, and redistribution on (non)renewable energy consumption. Beyond the interactive results, we follow the recommendation of Brambor et al. (2006) to apply the marginal effect analysis to examine how corruption, income inequality, and redistribution change in (non)renewable energy consumption across the geographical regions.[3] Table 5 presents the interactive effects on total, nonrenewable, and renewable energy consumption.

From Table 5, the estimates show that the interaction between SSA's dummy variable and post-tax/transfer Gini index (*SSA* × *lngini_disp)* on total energy consumption is positive and statistically significant (see Model 1). However, *SSA* × *lngini_disp* has a statistically significant negative effect on nonrenewable and renewable energy consumption (Models 5 & 9, respectively). The marginal impact of *SSA* × *lngini_disp* suggests that compared to Latin America and Caribbean economies (baseline category), for every 1% increase in post-tax/transfer Gini index, total energy consumption in SSA increases by 2.252 percentage points. Also, the marginal effect of *SSA* × *lngini_disp* suggests that compared to Latin America and Caribbean economies (baseline category), for every 1% increase in post-tax/transfer Gini index, nonrenewable and renewable energy consumption in SSA decreases by 4.452 and 0.045 percentage points, respectively. Also, the interaction between the East Asia and Pacific dummy variable and post-tax/transfer Gini index (*EAP* × *lngini_disp)* on total energy consumption and the nonrenewable energy consumption is negative and statistically significant (see Models 1 & 5). On the contrary, *EAP* × *lngini_disp* has a statistically significant positive effect on renewable energy consumption (see Model 9). The marginal effect analysis suggests that relative to Latin America and Caribbean economies, for every 1% increase in post-tax/transfer Gini index, total energy consumption and nonrenewable energy consumption in East Asia and the Pacific reduce by 5.436 and 7.659 percentage points, respectively. Contrarily, compared to Latin America and Caribbean economies, for every 1% increase in post-tax/transfer Gini index, renewable energy consumption in East Asia and the Pacific increases by 6.053 percentage points.

The interaction between the Middle East and North Africa dummy variable and post-tax/transfer Gini index *(MENA* × *lngini_disp*) has a positive effect on total energy consumption (see Model 1) and nonrenewable energy consumption (see Model 5), while its effects on renewable energy are negative (see Model 9). The marginal effect analysis suggests that relative to Latin America and Caribbean economies, for every 1% increase in post-tax/transfer Gini index, total energy consumption and nonrenewable energy consumption in the Middle East and North Africa increase by 1.85 and 4.319 percentage points, respectively. Contrarily, compared to Latin America and Caribbean economies, for every 1% increase in post-tax/transfer Gini index, renewable energy consumption in the Middle East and North Africa reduces by 1.198 percentage points. Similarly, the interaction between Europe and Central Asia dummy variable and post-tax/transfer Gini

[3] The marginal effects of income inequality can be using $\frac{\partial lnEC}{\partial GINI} = \beta_J + \delta_1 \times (\textit{regional dummy})$. Where β_J is the income inequality variables coefficients, δ_1 is the regional dummies and income inequality interaction terms coefficient. Regional dummies represent each geographical region, which should always be equal to one (1), when other regions are set to zero. The same approach should be followed when calculating the marginal effect of income redistribution. Following this approach, the marginal effect of income inequality for SSA is $\frac{\partial lnEC}{\partial GINI} = -1.434 + 3.686 \times (1) = 2.252$.

Table 5
Effect of income inequality, redistribution and corruption on energy nonrenewable and renewable energy consumption, including regional dummies with regional interactions.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
L.lnener	0.660***	0.731***	0.348***	0.398***								
	(0.007)	(0.010)	(0.116)	(0.128)								
L.lnfosil					0.599***	0.543***	0.696***	0.669***				
					(0.008)	(0.009)	(0.191)	(0.182)				
L.lnrenew									0.536***	0.550***	0.420**	0.642***
									(0.007)	(0.006)	(0.174)	(0.163)
lnrgdpc	0.110***	0.139***	0.127	0.045	0.203***	0.174***	−0.043	−0.030	0.127***	0.036	0.421*	0.485*
	(0.009)	(0.013)	(0.152)	(0.164)	(0.017)	(0.012)	(0.079)	(0.088)	(0.025)	(0.022)	(0.238)	(0.250)
lnfdi	−0.005***	−0.005***	0.009	0.017	−0.025***	−0.019***	0.007	0.009	0.003	0.001	−0.020	0.036
	(0.002)	(0.001)	(0.021)	(0.023)	(0.001)	(0.002)	(0.017)	(0.018)	(0.003)	(0.003)	(0.064)	(0.064)
lntrade	0.076***	0.072***	0.189**	0.155**	0.107***	0.081***	0.071	0.012	−0.233***	−0.218***	−0.376*	−0.562***
	(0.004)	(0.007)	(0.082)	(0.069)	(0.006)	(0.010)	(0.087)	(0.110)	(0.027)	(0.019)	(0.206)	(0.216)
lninfla	0.018***	0.021***	0.030	0.029	0.006***	0.005***	−0.003	−0.009	−0.022***	−0.020***	−0.039	−0.059
	(0.001)	(0.001)	(0.025)	(0.023)	(0.001)	(0.001)	(0.019)	(0.019)	(0.002)	(0.002)	(0.073)	(0.065)
lnoda	0.002***	0.010***	0.007	−0.000	0.005***	0.008***	0.020	0.012	0.017***	0.027***	−0.119*	−0.158**
	(0.001)	(0.001)	(0.034)	(0.030)	(0.001)	(0.001)	(0.019)	(0.020)	(0.004)	(0.003)	(0.064)	(0.066)
lnsse	−0.052***	−0.057***	0.070	0.167	0.219***	0.243***	−0.105	−0.244	−0.145***	0.005	0.033	−0.101
	(0.006)	(0.008)	(0.367)	(0.282)	(0.020)	(0.012)	(0.260)	(0.278)	(0.036)	(0.024)	(0.577)	(0.579)
corrup	0.069***	0.085***	−0.039	−0.045	−0.036***	−0.004	0.059	0.037	−0.008	−0.042***	−0.302	−0.469*
	(0.007)	(0.008)	(0.158)	(0.146)	(0.006)	(0.006)	(0.079)	(0.083)	(0.014)	(0.011)	(0.282)	(0.245)
SSA	−14.108***	−8.472***	−19.672	−15.861	19.873***	17.756***	0.312	−5.705	7.242***	9.227***	−17.467	−9.883
	(0.404)	(0.708)	(46.217)	(44.718)	(0.807)	(0.745)	(19.263)	(15.672)	(0.738)	(1.161)	(70.960)	(69.528)
EAP	13.858***	15.693***	1.494	0.725	30.487***	18.002***	0.991	1.375	−16.626***	4.013	−9.681*	−6.410*
	(4.490)	(4.061)	(1.402)	(0.999)	(3.964)	(4.496)	(0.695)	(0.854)	(4.471)	(4.180)	(5.610)	(3.661)
MENA	−11.756***	−14.898***	4.885	−0.803	−12.426***	−10.453***	5.432	5.212	10.397***	17.419***	−21.031**	−11.226**
	(1.779)	(1.819)	(7.504)	(4.404)	(2.294)	(1.698)	(5.095)	(3.931)	(1.972)	(1.742)	(9.814)	(5.609)
ECA	−4.592***	−2.989***	−1.131	−1.145	2.243***	7.356***	0.970	0.604	−3.749**	−6.342***	1.650	2.785
	(0.389)	(0.679)	(1.450)	(0.964)	(0.543)	(0.589)	(1.063)	(1.064)	(1.862)	(2.147)	(2.547)	(2.502)
SAR	−9.865***	21.146***	0.279	−0.074	21.875***	25.608***	−0.706	−0.822	−90.794***	−48.013***	8.603	7.247
	(3.194)	(4.518)	(0.930)	(0.783)	(4.429)	(3.331)	(2.174)	(1.294)	(22.986)	(14.305)	(7.793)	(5.551)
lngini_disp	−1.434***				0.724***				1.514***			
	(0.096)				(0.133)				(0.263)			
SSA × lngini_disp	3.686***				−5.176***				−1.559***			
	(0.112)				(0.218)				(0.200)			
EAP × lngini_disp	−4.002***				−8.383***				4.539***			
	(1.255)				(1.089)				(1.218)			
MENA × lngini_disp	3.284***				3.595***				−2.712***			
	(0.485)				(0.637)				(0.544)			
ECA × lngini_disp	1.239***				−0.497***				1.266**			
	(0.102)				(0.143)				(0.520)			
SAR × lngini_disp	2.559***				−5.957***				25.027***			
	(0.875)				(1.215)				(6.213)			
lngini_mkt		−0.877***				0.829***				2.044***		
		(0.160)				(0.141)				(0.314)		
SSA × lngini_mkt		2.161***				−4.507***				−2.109***		
		(0.169)				(0.188)				(0.304)		
EAP × lngini_mkt		−4.273***				−4.947***				−1.001		
		(1.095)				(1.218)				(1.097)		
MENA × lngini_mkt		4.022***				2.986***				−4.687***		
		(0.487)				(0.438)				(0.457)		
ECA × lngini_mkt		0.821***				−1.839***				1.729***		
		(0.173)				(0.158)				(0.565)		
SAR × lngini_mkt		−5.718***				−6.849***				12.898***		

(*continued on next page*)

Table 5 (*continued*)

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
		(1.201)				(0.917)				(3.804)		
lnrel_red			0.309***				0.120**				−0.428**	
			(0.108)				(0.049)				(0.214)	
SSA × lnrel_red			8.987				0.018				6.199	
			(18.577)				(7.023)				(27.440)	
EAP × lnrel_red			−0.466**				−0.165				0.740**	
			(0.237)				(0.111)				(0.301)	
MENA × lnrel_red			−2.555				−2.190				6.255*	
			(3.694)				(2.607)				(3.716)	
ECA × lnrel_red			0.300				−0.381				−0.267	
			(0.411)				(0.327)				(0.796)	
SAR × lnrel_red			−0.483				−0.021				−1.590	
			(1.555)				(1.211)				(2.340)	
lnabs_red				0.285***				0.151**				−0.288
				(0.075)				(0.075)				(0.198)
SSA × lnabs_red				8.971				3.212				5.101
				(21.268)				(6.963)				(31.348)
EAP × lnabs_red				−0.379*				−0.228				0.502**
				(0.219)				(0.139)				(0.254)
MENA × lnabs_red				0.120				−3.493				3.540
				(3.457)				(3.122)				(3.028)
ECA × lnabs_red				0.453				−0.356				−0.778
				(0.377)				(0.428)				(1.067)
SAR × lnabs_red				−0.305				−0.514				−2.688
				(1.481)				(1.272)				(3.650)
Constant	6.748***	3.797***	1.655	2.078*	−4.390***	−4.436***	1.269	2.336	−4.489***	−6.714***	2.695	2.957
	(0.416)	(0.713)	(1.486)	(1.064)	(0.601)	(0.620)	(1.204)	(1.545)	(1.067)	(1.322)	(2.049)	(2.342)
Observations	851	851	398	398	812	812	398	398	1058	1058	421	421
Hansen	80.708	73.227	7.574	8.527	63.898	60.100	6.055	6.060	91.918	83.357	10.010	10.521
P(Hansen)	0.520	0.745	0.670	0.577	0.931	0.967	0.811	0.810	0.366	0.620	0.440	0.396
AR (1)	0.000	0.000	0.046	0.039	0.024	0.024	0.149	0.190	0.043	0.023	0.089	0.056
AR (2)	0.352	0.271	0.960	0.936	0.382	0.520	0.696	0.935	0.215	0.217	0.892	0.768
No. of countries	166	166	166	166	166	166	166	166	166	166	166	166
No. of instruments	102	102	30	30	102	102	30	30	108	108	30	30

Standard errors in parentheses. Hansen-test refers to the over-identification test for the restrictions in system-GMM estimation. The AR (1) and AR (2) tests are the Arellano–Bond tests for first and second-order autocorrelation in first differences.

* $p < 0.10$,.

** $p < 0.05$,.

*** $p < 0.01$.

Note: For regional Comparison, Latin America and the Caribbean region is used as the baseline. SSA= sub-Saharan Africa, EAP = East Asia & Pacific, MENA = Middle East and North Africa, ECA =Europe and Central Asia and SAR =South Asia.

index (*ECA* × *lngini_disp*) has a positive effect on total energy consumption (see Model 1) and renewable energy consumption (see Model 9), whereas its effects on nonrenewable energy consumption are negative. Thus, the marginal effect analysis shows that for every 1% increase in post-tax/transfer Gini index, total energy consumption reduces by 0.195 percentage points while nonrenewable and renewable energy consumption increases by 0.227 and 2.78 percentage points, respectively, compared to Latin America & Caribbean countries. Also, the interaction between South Asia and post-tax/transfer Gini index (*SAR* × *lngini_disp*) has a positive effect on total energy consumption (see Model 1) and renewable energy consumption (see Model 9), while it affects nonrenewable energy consumption negatively. The marginal effect analysis shows that for every 1% increase in post-tax/transfer Gini index, total energy consumption and renewable energy consumption in South Asia increases by 1.125 and 26.541 percentage point, respectively, while nonrenewable energy consumption decreases by 5.233 percentage point compared to Latin America & Caribbean countries.

The interaction between the SSA dummy and the pre-tax/transfer Gini index (*SSA* × *lngini_mkt)* has a significant positive effect on total energy consumption (see Model 2). However, the impact of *SSA* × *lngini_mkt* on nonrenewable energy (see Model 6) and renewable energy consumption (see Model 10) is negative and statistically significant. The marginal effect analysis indicates that relative to Latin America & Caribbean, for every 1% increase in pre-tax/transfer Gini index, SSA economies' total energy consumption increases by 1.284 percentage points, respectively, while nonrenewable and renewable energy consumption reduces by 3.678 and 0.065 percentage points respectively. Also, the interaction between East Asia and Pacific dummy variable and pre-tax/transfer Gini index (*EAP* × *lngini_mkt*) reduces total energy consumption (see Model 2), nonrenewable energy consumption (see Model 6) and has an insignificant effect on renewable energy consumption (see Model 10). The marginal effect analysis suggests that relative to Latin America & Caribbean, for every 1% increase in pre-tax/transfer Gini index, East Asia and Pacific region's total energy and nonrenewable energy consumption decreases by 5.15 and 4.118 percentage points. Also, the interaction between the Middle East and North Africa dummy variable and pre-tax/transfer Gini index (*MENA* × *lngini_mkt*) has a positive effect on total energy consumption (see Model 2) and nonrenewable energy consumption (see Model 6) while its impact on renewable energy consumption is negative and statistically significant (see Model 10). Compared to Latin America and the Caribbean region, the marginal effect analysis shows that for every 1% increase in pre-tax/transfer Gini index, the MENA region's total energy and nonrenewable energy consumption increases by 3.145 and 3.815 percentage points, respectively, while renewable energy consumption decreases by 2.643 percentage point.

In addition, the interaction between Europe and Central Asia dummy variable and pre-tax/transfer Gini index (*ECA* × *lngini_mkt*) has a significant positive effect on total energy consumption (see Model 2) and renewable energy consumption (see Model 10), whereas its effects on nonrenewable energy (see Model 6) consumption is negative and statistically significant. Relative to Latin America and the Caribbean region, the marginal effect analysis indicates that for every 1% increase in pre-tax/transfer Gini index, Europe and Central Asia's total energy and nonrenewable energy consumption decreases by 0.056 and 1.01 percentage points, respectively, while renewable energy consumption increases by 3.773 percentage point. The interaction between South Asia and pre-tax/transfer Gini index (*SAR* × *lngini_mkt*) has a significant negative effect on total energy consumption (see Model 2) and nonrenewable energy consumption (see Model 6), while its effects on renewable energy consumption (see Model 10) is positive and statistically significant. Relative to Latin America and the Caribbean region, marginal effect analysis indicates that for every 1% increase in pre-tax/transfer Gini index, total energy and nonrenewable energy consumption in South Asia decreases by 6.595 and 6.02 percentage points, respectively, while renewable energy consumption increases by 14.942 percentage point.

These analyses suggest that the effect of income inequality on the energy mix differs across regions. For instance, in SSA, post-tax/transfer and pre-tax/transfer income inequality increase total energy consumption while reducing nonrenewable and renewable energy consumption. Sub-Saharan Africa is one of the most unequal regions in the world (Bhorat et al., 2017). This result supports the notion that higher income inequality could reduce the pace of renewable energy usage in SSA. At the same time, the higher income inequality in SSA is important for minimizing nonrenewable energy usage. For East Asia and the Pacific, post-tax/transfer and pre-tax/transfer income inequality reduce total and nonrenewable energy consumption while increasing renewable energy consumption. This result reflects the significant drop in income inequality in the EAP region. Thus, the continuous decline in income inequality in EAP has promoted energy efficiency by reducing the region's primary energy consumption and overreliance on non-renewable energy. Also, the relatively low income inequality has been crucial for facilitating renewable energy usage in EAP.

In the MENA region, post-tax/transfer and pre-tax/transfer income inequality increase total and nonrenewable energy consumption while reducing renewable energy consumption. MENA region has been identified as an unequal region in the world, with the top 10% income share around 58% and the bottom 50% income share around 8% (Chancel et al., 2022). This implies that the higher income inequality in MENA contributes significantly to excessive primary and nonrenewable energy use. At the same time, the higher income inequality in the region is a major hindrance to transitioning towards a renewable energy economy. On the other hand, the analysis emphasises that post-tax/transfer and pre-tax/transfer income inequality reduce total energy consumption while increasing ECA's nonrenewable and renewable energy consumption. This result is likely due to the relatively lower income disparity within the ECA region. Europe is the most equal region in the world, with the top 10% income share around 36% and the bottom 50% income share around 18% (Chancel et al., 2022). At the same time, greater income equality exists in Central Asia countries (Kidd et al., 2022). We suggest, therefore, that the relatively low income inequality in ECA has been important for supporting the region's transition towards a renewable energy economy while improving energy efficiency by reducing primary energy consumption. In South Asia, post-tax/transfer income inequality increases total and renewable energy consumption while reducing nonrenewable energy consumption. At the same time, the pre-tax/transfer Gini index reduces total and nonrenewable energy consumption while increasing nonrenewable energy consumption in South Asia.

Regarding the interaction between the regional dummy variables and income redistribution, the estimates show that the interaction between *SSA* × *relative income redistribution (lnrel_red) has an insignificant* positive effect on total energy consumption (see Model 3), nonrenewable energy (see Model 7) and renewable energy consumption (see Model 11). The interaction between *EAP* × *lnrel red* has a significant negative effect on total energy consumption (see Model 3) and a significant positive effect on renewable energy consumption (see Model 11). The interaction between *EAP* × *lnrel red* has an insignificant impact on nonrenewable energy consumption (see Model 7). Relative to Latin America and the Caribbean region, the marginal effect analysis indicates that for every 1% increase in relative income redistribution, total energy consumption in EAP reduces by 0.157 percentage points while renewable energy consumption increases by 0.312 percentage points. The interaction between *MENA* × *lnrel red* has an insignificant negative effect on total energy consumption (see Model 3) and nonrenewable energy consumption (see Model 7) while exerting a significant positive impact on renewable energy (see Model 11). Thus, relative to Latin America and the Caribbean region, a 1% increase in relative income redistribution in MENA increases renewable energy consumption by 5.827 percentage points. It is observed that the interaction between *ECA* × *lnrel red* has an insignificant effect on total energy consumption (see

Table 6
Effect of control of corruption on energy nonrenewable and renewable energy consumption, including regional dummies with regional interactions.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
L.lnener	0.688***	0.727***	0.303***	0.343***								
	(0.011)	(0.010)	(0.074)	(0.072)								
L.lnfosil					0.719***	0.727***	0.902***	0.867***				
					(0.007)	(0.008)	(0.198)	(0.196)				
L.lnrenew									0.661***	0.700***	0.436**	0.463**
									(0.013)	(0.013)	(0.215)	(0.220)
lnrgdpc	0.241***	0.238***	0.172*	0.150	0.036***	0.017**	−0.098	−0.135	−0.121***	−0.124***	−0.320	−0.411
	(0.011)	(0.011)	(0.090)	(0.110)	(0.009)	(0.007)	(0.128)	(0.124)	(0.017)	(0.014)	(0.512)	(0.570)
lnfdi	0.001	−0.000	0.012	0.009	−0.003***	−0.004***	−0.006	−0.003	0.004	0.000	0.016	0.019
	(0.001)	(0.001)	(0.008)	(0.009)	(0.001)	(0.001)	(0.016)	(0.016)	(0.003)	(0.003)	(0.042)	(0.043)
lntrade	0.031***	0.023**	0.112***	0.103**	0.048***	0.049***	0.075	0.088	−0.097***	−0.062***	0.056	−0.001
	(0.008)	(0.009)	(0.042)	(0.041)	(0.006)	(0.007)	(0.078)	(0.073)	(0.015)	(0.017)	(0.228)	(0.268)
lninfla	0.020***	0.019***	0.019***	0.022***	0.006***	0.006***	0.006	0.008	−0.029***	−0.027***	−0.088*	−0.093*
	(0.001)	(0.001)	(0.007)	(0.007)	(0.000)	(0.001)	(0.017)	(0.016)	(0.002)	(0.001)	(0.046)	(0.051)
lnoda	0.011***	0.014***	−0.003	−0.005	0.013***	0.016***	0.033*	0.037**	0.038***	0.036***	−0.003	−0.002
	(0.001)	(0.001)	(0.014)	(0.015)	(0.001)	(0.001)	(0.019)	(0.015)	(0.003)	(0.003)	(0.076)	(0.082)
lnsse	−0.153***	−0.133***	0.142	0.180	0.231***	0.249***	−0.068	0.014	−0.074***	−0.024	0.802	0.742
	(0.011)	(0.013)	(0.119)	(0.152)	(0.012)	(0.010)	(0.260)	(0.240)	(0.022)	(0.018)	(0.733)	(0.746)
corrup	−0.034***	−0.054***	−0.192**	−0.224***	0.027	0.061***	−0.035	−0.022	0.084***	0.100***	0.408	0.303
	(0.012)	(0.014)	(0.092)	(0.071)	(0.018)	(0.015)	(0.106)	(0.084)	(0.031)	(0.026)	(0.387)	(0.526)
SSA	0.005	0.106	1.201	1.638	0.027	−0.033	−1.035	−1.001	−0.000	−0.016	−0.801	−0.844
	(0.065)	(0.067)	(1.105)	(1.033)	(0.030)	(0.024)	(0.785)	(0.807)	(0.060)	(0.052)	(2.104)	(2.335)
EAP	−0.103	0.016	0.097	0.144	0.145***	0.075	0.150	0.164	−0.016	−0.104	−0.616	0.467
	(0.168)	(0.137)	(0.505)	(0.402)	(0.048)	(0.061)	(0.821)	(0.735)	(0.143)	(0.106)	(2.714)	(2.624)
MENA	0.319***	0.244**	−0.178	−0.175	−0.020	−0.082***	1.041	1.070	−0.803***	−0.786***	−2.313	−1.590
	(0.109)	(0.110)	(0.853)	(0.699)	(0.034)	(0.029)	(0.768)	(0.812)	(0.167)	(0.151)	(3.369)	(3.887)
ECA	0.147**	0.156***	0.194	0.254**	−0.069**	−0.158***	−0.207	−0.222	−0.010	−0.046	0.066	0.009
	(0.058)	(0.052)	(0.124)	(0.113)	(0.033)	(0.025)	(0.156)	(0.151)	(0.056)	(0.039)	(0.443)	(0.462)
SAR	−0.127	−0.184*	−0.670	−0.094	0.033	−0.028	−0.518	−0.664	0.082	0.094	−0.551	−1.623
	(0.125)	(0.110)	(0.945)	(0.960)	(0.051)	(0.049)	(1.011)	(1.001)	(0.144)	(0.121)	(4.681)	(4.908)
SSA × corrup	0.233***	0.259***	0.528*	0.483	−0.030	−0.074***	0.361	0.202	−0.238***	−0.245***	−0.999	−1.153
	(0.025)	(0.027)	(0.296)	(0.343)	(0.023)	(0.020)	(1.149)	(1.008)	(0.060)	(0.053)	(2.729)	(3.019)
EAP × corrup	0.159***	0.190***	−0.563	−1.039**	0.125***	0.051	−1.044	−0.907	−0.342***	−0.400***	3.867	3.936
	(0.018)	(0.024)	(0.489)	(0.491)	(0.031)	(0.035)	(1.539)	(1.760)	(0.051)	(0.048)	(4.614)	(4.663)
MENA × corrup	0.622***	0.709***	−0.732	−0.744	0.094**	0.071***	−0.141	−0.235	−0.453***	−0.562***	−3.413**	−3.084*
	(0.014)	(0.021)	(0.676)	(0.591)	(0.043)	(0.024)	(0.642)	(0.542)	(0.073)	(0.072)	(1.512)	(1.664)
ECA × corrup	−0.208***	−0.177***	−0.026	0.018	−0.075***	−0.102***	0.123	0.157	0.272***	0.199***	−0.314	−0.179
	(0.017)	(0.018)	(0.124)	(0.106)	(0.018)	(0.018)	(0.119)	(0.116)	(0.038)	(0.032)	(0.713)	(0.891)
SAR × corrup	−0.569***	−0.637***	−0.386	0.537	−0.084**	−0.097**	1.984	1.898	0.389***	0.459***	−5.939	−4.988
	(0.124)	(0.127)	(0.575)	(0.715)	(0.033)	(0.039)	(2.657)	(2.825)	(0.137)	(0.158)	(10.522)	(10.139)
lngini_disp	−0.159**				0.202***				0.129*			
	(0.065)				(0.062)				(0.074)			
lngini_mkt		0.085				0.023				0.255***		
		(0.117)				(0.068)				(0.077)		
lnrel_red			0.176***				0.108				−0.421	
			(0.048)				(0.068)				(0.258)	
lnabs_red				0.158***				0.104				−0.358
				(0.049)				(0.076)				(0.295)
Constant	0.969***	−0.314	1.931***	1.872***	−1.341***	−0.618**	0.414	0.471	1.634***	0.728	1.913	2.686
	(0.319)	(0.551)	(0.573)	(0.558)	(0.280)	(0.265)	(0.978)	(1.028)	(0.416)	(0.449)	(3.020)	(3.388)
Observations	851	851	398	398	812	812	398	398	1058	1058	421	421
Hansen	82.139	78.877	18.784	18.414	70.388	73.169	6.682	6.657	91.304	90.118	3.039	3.034
P(Hansen)	0.475	0.577	0.845	0.860	0.816	0.746	0.755	0.757	0.384	0.417	0.804	0.805

(*continued on next page*)

Table 6 (*continued*)

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
AR (1)	0.000	0.000	0.109	0.097	0.008	0.008	0.038	0.045	0.011	0.012	0.334	0.233
AR (2)	0.210	0.209	0.206	0.132	0.681	0.681	0.240	0.291	0.259	0.280	0.107	0.114
No. of countries	166	166	166	166	166	166	166	166	166	166	166	166
No. of instruments	102	102	46	46	102	102	30	30	108	108	26	26

Standard errors in parentheses. Hansen-test refers to the over-identification test for the restrictions in system-GMM estimation. The AR (1) and AR (2) tests are the Arellano–Bond tests for first and second-order autocorrelation in first differences.

$^{*}p < 0.10$,.

$^{**}p < 0.05$,.

$^{***}p < 0.01$.

Note: For regional Comparison, Latin America and the Caribbean region is used as the baseline. SSA= sub-Saharan Africa, EAP = East Asia & Pacific, MENA = Middle East and North Africa, ECA =Europe and Central Asia and SAR =South Asia.

Model 3), nonrenewable energy (see Model 7) consumption, and renewable energy consumption (see Model 11). Similar results are observed in the interaction between *SAR* × *lngini_mkt* on total energy consumption (see Model 3), nonrenewable energy consumption (see Model 7), and renewable energy consumption (see Model 11).

Again, the interaction effect of *SSA* × *absolute income redistribution* (*lnabs_red*) shows an insignificant effect on total energy consumption (see Model 4), nonrenewable energy (see Model 8), and renewable energy consumption (see Model 12). The interaction between *EAP* × *lnabs_red* has a significant negative effect on total energy consumption (see Model 4), an insignificant effect on nonrenewable energy consumption (see Model 8), and a significant positive effect on renewable energy consumption (see Model 12). Thus, relative to Latin America and the Caribbean region, a 1% increase in absolute income redistribution in EAP decreases total energy consumption by 0.094 percentage points while increasing renewable energy consumption by 0.214 percentage points. Finally, the interaction between *MENA* × *lnabs_red, ECA* × *lnabs_red, and* SAR × *lnabs_red* have an insignificant effect on total energy, nonrenewable and renewable energy consumption.

The income redistribution analysis shows the effect of relative and absolute income redistribution on the energy mix is only significant in a few regions. For instance, our analysis depicts that in EAP, relative and absolute income redistribution significantly reduces total energy consumption while increasing renewable energy consumption. In MENA, relative income redistribution significantly increases renewable energy consumption. In the remaining regions, we observed that both absolute and relative income redistribution does not affect total, nonrentable and renewable energy consumption. Given the income inequality level in EAP and MENA regions, these findings highlight that pursuing income redistribution policies to address income inequality would substantially support these regions' renewable energy transition agenda.

We present the interactive effect of corruption and the regional dummy variables in Table 6. For the interaction analysis, *SSA* × *corrup* has a significant positive effect on total energy consumption (see Model 1–3), whereas its effects on nonrenewable energy (see Model 6) and renewable energy consumption (see Model 9 & 10) are negative. Also, the interaction between *EAP* × *corrup* has a significant positive effect on total energy consumption (see Models 1 & 2) and nonrenewable energy consumption (see Model 5), whereas its impact on renewable energy (see Model 12) is negative. The interaction between *MENA* × *corrup* has a significant positive effect on total energy consumption (see Models 1 & 2) and nonrenewable energy consumption (see Models 5 & 6) while having a significant negative effect on renewable energy consumption (see Models 9–12). These results indicate that the current efforts to address corruption in SSA, MENA and EAP regions are insufficient to support energy transition and address energy efficiency. For instance, corruption control has been minimal in SSA, posing a challenge in the renewable energy sector (Acheampong et al., 2021). On the other hand, the interaction between *ECA* × *corrup* has a significant negative effect on total energy consumption (see Models 1 & 2) and nonrenewable energy consumption (see Models 5 & 6), while it has a significant positive effect on renewable energy consumption (see Models 9 & 10). Similarly, the interaction between *SAR* × *corrup* has a significant negative effect on total energy consumption (see Models 1 & 2), non-renewable energy consumption (see Models 5 & 6), while it impacts renewable energy consumption (see Models 9 & 10) is positive and statistically significant. These results suggest that in ECA and SAR, the extent of corruption control has effectively minimised nonrenewable energy consumption while enhancing energy efficiency and renewable energy usage in their economy.

4.4. Examining if corruption conditions the effect of income inequality and redistribution on (non)renewable energy consumption

From Eqs. (2)-(3), we present the results if corruption conditions income inequality and redistribution to affect (non)renewable energy consumption.[4] Therefore, Table 7 shows the interaction between corruption, income inequality, and redistribution on nonrenewable, renewable, and total energy consumption. Corruption reduces environmental regulations' strictness and makes energy policies inefficient (Fisman & Svensson, 2007). This distorts public expenditure intended for other purposes, such as poverty and income inequality alleviation (see Salinas-Jimenez and Salinas-Jimenez, 2007; Wu et al., 2010) and, thus, affects the energy demand.

From Table 7, the interaction between control of corruption and post-tax/transfer Gini index (*corrup* × *lngini_disp*) has a significant positive effect on total energy consumption (see Model 1) and nonrenewable energy (see Model 6), whereas it impacts renewable energy consumption (see Model 9) is negative and statistically significant. From marginal effect analysis,[5] when control of corruption increases, the post-tax/transfer Gini index increases total energy and nonrenewable energy consumption by 0.909 and 0.368 percentage points, respectively, while renewable energy consumption reduces by 0.185 percentage points. The implication is that when corruption is effectively controlled, income inequality (post-tax/transfer income inequality) would still contribute energy inefficiency through the consumption of non-renewable (fossil) energy and the reduction of renewable energy consumption. We suggest that when corruption control ensures that tax revenues are used to support energy transition policies and subsidize energy usage by poorer households, it would drive overall energy efficiency, reduce nonrenewable energy usage, and increase renewable energy consumption. In addition, the interaction between control of corruption and pre-tax/transfer Gini index (*corrup* × *lngini_mkt*) has a significant positive effect on total energy consumption (see Model 2) and nonrenewable energy consumption (see Model 6) and renewable energy (see Model 10). The marginal analysis shows that when control of corruption increases, the pre-tax/transfer Gini index increases total energy, nonrenewable, and renewable energy consumption by 0.921, 0.834, and 0.180 percentage points, respectively. The implication is that when corruption is managed effectively, income inequality before tax will improve energy efficiency through renewable energy consumption. At the same time, even when corruption is managed, income inequality before tax contributes to reducing non-renewable energy consumption.

Regarding the interaction between control of corruption and income redistribution, Table 7 shows that the interaction between control of corruption and relative income redistribution (*corrup* × *lnrel_red*) has a significant positive effect on renewable energy (see Model 12) while it has an insignificant effect on total energy (see Model 3) and statistically significant negative effect on nonrenewable energy consumption (see Model 9). Thus, from the marginal analysis, when control of corruption increases, relative income redistribution reduces nonrenewable energy consumption by 0.146 percentage points while increasing renewable energy consumption by 0.436 percentage points. Also, the interaction between control of corruption and absolute income redistribution (*corrup* × *lnabs_red*) has an insignificant negative effect on total energy consumption (see Model 4), a significant negative effect on nonrenewable energy consumption (see Model 8), and a statistically significant positive effect on renewable energy consumption (see Model 12). Thus, from the marginal analysis, when control of corruption increases, absolute income redistribution reduces nonrenewable energy consumption by 0.117 percentage points while increasing renewable energy consumption by 0.482 percentage points. The policy relevance of these results is that with control of corruption, both relative and absolute income redistribution would directly increase poor households' income. Increasing poorer household income enables these households to transition towards renewable energy relative to nonrenewable energy consumption. In other words, we argue that when income is redistributed directly to the households, it may increase their disposable income. Such income may be used to purchase renewable energy services relative to non-renewable energy. Therefore, we argue that when corruption control supports income redistribution to drive renewable energy transition, policymakers should use the tax revenue to support and subsidize renewable energy usage by poorer households.

4.5. Robustness check with different econometric estimators

In this section, we deploy different econometric techniques to test the robustness of our findings. We used the instrumental variable generalized method of moment estimator (IV-GMM) to further control for endogeneity, Driscol-Kraay to control cross-sectional and temporal dependency and Feasible Generalized Least Square (FGLS) for controlling heteroskedasticity, cross-sectional and serial correlations. The results presented in Appendix Table A3 align with the previous findings discussed. For instance, across the IV-GMM, Driscol-Kraay and FGLS models, post-tax/transfer Gini Index (lngini_disp) has a statistically significant and negative effect on total energy consumption. Similarly, the pre-tax/transfer Gini Index (lngini_mkt) has a statistically significant and negative effect on total energy consumption across all econometric techniques. Regarding income redistribution, relative income redistribution (lnrel_red) has a statistically significant and positive effect on total energy consumption across all the econometric techniques. Similarly, absolute income redistribution (lnabs_red) has a statistically significant and positive effect on total energy consumption. Also, across all the econometric techniques, control of corruption has a negative and statistically significant effect on total energy consumption. The agreement amongst all the econometric techniques regarding the impact of income inequality, redistribution and control of corruption on energy mix suggests that our findings are robust and reliable for informing policies seeking to improve energy efficiency and transition across the globe.

5. Conclusion and policy implications

In this study, we argue that achieving the objectives of the Conference of the Parties (COP) 26 regarding net-zero and energy transition depends on the efficacy of institutions and income redistribution policies. To support our conjecture, we provide empirical evidence on the impact of corruption, income inequality, and redistribution on renewable and nonrenewable and total energy consumption using panel data from 166 countries. In alignment with the research objective, we seek to answer the following questions, which also answer our research hypothesis: (1) *Do corruption, income inequality, and redistribution contribute to over-dependence on nonrenewable energy use? (2) Do corruption, income inequality, and redistribution hinder the transition to a renewable energy*

[4] The predictive margins of the interactive results for the aggregated sample are display in the Fig. A1, Fig. A2 and Fig. A3. The predictive margins results are consistent with the interpretation of the interactive and marginal effect results.

[5] We calculated the marginal effect of control of corruption conditioned on income inequality using the equation $\frac{\partial lnEC}{\partial lnGINI} = \beta_2 + \delta_1 CORRUP$. Where β_2 is the coefficient of income inequality variable in equation (2), δ_1 is the control of corruption and income inequality interaction terms coefficient. In this study, we computed the marginal effect of income inequality conditioned on the maximum value (2.470) of control of corruption (see Table 2). For instance, following this approach the marginal effect for Model 1 in Table 7 can be computed as $\frac{\partial lnEC}{\partial lnGINI} = -0.161 + 0.433(2.470) = 0.909$. The same approach can be followed to compute the marginal effect control of corruption conditioned on income inequality for the other models in Table 7. Also, the same approach $\left(\frac{\partial lnEC}{\partial lnREDIST} = \beta_3 + \delta_2 CORRUP\right)$, Where β_3 is the coefficient of income redistribution variable in equation (3), δ_2 is the control of corruption and income inequality interaction terms coefficient can be followed to compute the marginal effect of income redistribution conditioned on control of corruption.

Table 7
Interactions between control of corruption, income inequality and redistribution on energy nonrenewable and renewable energy consumption.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
	Total energy consumption				Nonrenewable energy consumption				Renewable energy consumption			
L.lnener	0.830***	0.870***	0.677***	0.706***								
	(0.003)	(0.004)	(0.050)	(0.053)								
L.lnfosil					0.784***	0.766***	0.992***	1.006***				
					(0.003)	(0.002)	(0.034)	(0.034)				
L.lnrenew									0.809***	0.804***	0.930***	0.935***
									(0.003)	(0.002)	(0.023)	(0.023)
lnrgdpc	0.088***	0.033***	0.091	0.061	0.045***	0.056***	0.060***	0.064***	−0.061***	−0.074***	0.074*	0.067
	(0.002)	(0.004)	(0.083)	(0.085)	(0.001)	(0.002)	(0.021)	(0.021)	(0.005)	(0.006)	(0.040)	(0.045)
lnfdi	−0.002***	0.001***	−0.028***	−0.029***	−0.005***	−0.005***	−0.002	−0.001	0.011***	0.012***	0.023***	0.025***
	(0.001)	(0.000)	(0.008)	(0.009)	(0.000)	(0.001)	(0.003)	(0.003)	(0.001)	(0.001)	(0.009)	(0.008)
lntrade	0.027***	0.013***	0.185***	0.186***	0.037***	0.031***	0.020**	0.013	−0.084***	−0.099***	−0.096***	−0.101***
	(0.003)	(0.004)	(0.034)	(0.034)	(0.002)	(0.003)	(0.010)	(0.010)	(0.005)	(0.006)	(0.030)	(0.031)
lninfla	0.010***	0.012***	0.021***	0.024***	0.003***	0.001***	0.002	0.001	−0.013***	−0.014***	−0.005	−0.004
	(0.000)	(0.000)	(0.006)	(0.007)	(0.000)	(0.000)	(0.003)	(0.003)	(0.001)	(0.001)	(0.007)	(0.006)
lnoda	0.005***	0.003***	−0.010	−0.010	0.015***	0.013***	0.010***	0.011***	0.007***	0.008***	0.002	−0.001
	(0.000)	(0.000)	(0.015)	(0.015)	(0.001)	(0.000)	(0.003)	(0.003)	(0.001)	(0.001)	(0.009)	(0.010)
lnsse	0.041***	0.067***	0.241	0.310	0.125***	0.130***	−0.159**	−0.185**	−0.168***	−0.163***	−0.359***	−0.389***
	(0.005)	(0.004)	(0.169)	(0.191)	(0.003)	(0.002)	(0.066)	(0.073)	(0.009)	(0.011)	(0.114)	(0.129)
corrup	−1.602***	−1.543***	0.054	0.079	−0.297***	−0.883***	0.159***	0.107***	0.357***	−0.440***	−0.326***	−0.244***
	(0.033)	(0.059)	(0.139)	(0.113)	(0.025)	(0.045)	(0.038)	(0.029)	(0.116)	(0.116)	(0.118)	(0.076)
lngini_disp	−0.161***				0.165***				−0.005			
	(0.020)				(0.015)				(0.019)			
corrup × lngini_disp	0.433***				0.082***				−0.073**			
	(0.010)				(0.007)				(0.031)			
lngini_mkt		−0.104**				0.293***				−0.181***		
		(0.045)				(0.023)				(0.026)		
corrup × lngini_mkt		0.415***				0.219***				0.146***		
		(0.016)				(0.011)				(0.030)		
lnrel_red			0.060				−0.025***				0.093***	
			(0.043)				(0.009)				(0.023)	
corrup × lnrel_red			−0.053				−0.049***				0.139***	
			(0.050)				(0.014)				(0.048)	
lnabs_red				0.020				−0.016**				0.114***
				(0.042)				(0.008)				(0.022)
corrup × lnabs_red				−0.073				−0.041***				0.149***
				(0.054)				(0.013)				(0.042)
Constant	0.650***	0.603***	−0.214	−0.369	−1.066***	−1.565***	0.016	0.020	2.036***	2.885***	1.238**	1.517***
	(0.075)	(0.209)	(0.508)	(0.490)	(0.068)	(0.093)	(0.103)	(0.104)	(0.115)	(0.129)	(0.482)	(0.544)
Observations	851	851	398	398	812	812	398	398	1058	1058	421	421
Hansen	91.083	87.688	18.713	17.416	73.179	78.891	15.817	16.385	97.598	99.415	20.807	21.447
P(Hansen)	0.478	0.579	0.475	0.562	0.914	0.813	0.669	0.631	0.464	0.413	0.348	0.313
AR (1)	0.000	0.000	0.001	0.001	0.007	0.009	0.009	0.008	0.015	0.015	0.028	0.028
AR (2)	0.579	0.592	0.113	0.171	0.889	0.810	0.710	0.685	0.509	0.495	0.781	0.772
No. of countries	166	166	166	166	166	166	166	166	166	166	166	166
No. of instruments	102	102	30	30	102	102	30	30	108	108	30	30

Standard errors in parentheses. Hansen-test refers to the over-identification test for the restrictions in system-GMM estimation. The AR (1) and AR (2) tests are the Arellano–Bond tests for first and second-order autocorrelation in first differences.

* $p < 0.10$,.

** $p < 0.05$,.

*** $p < 0.01$.

Note: For regional Comparison, Latin America and the Caribbean region is used as the baseline. SSA= sub-Saharan Africa, EAP = East Asia & Pacific, MENA = Middle East and North Africa, ECA =Europe and Central Asia and SAR =South Asia.

economy? (3) *Do corruption, income inequality, and redistribution explain the variation in (non)renewable energy consumption between geographical regions?* We deployed the two-step dynamic system-generalized method of moment as the estimation technique because of its capability of controlling endogeneity. The findings that emerged from this study are summarised as follows:

Our findings showed that net (post-tax/transfer Gini index) and market (pre-tax/transfer Gini index) income inequality reduces total and renewable energy consumption while increasing nonrenewable energy consumption. Also, our results indicated that absolute and relative income redistribution increases total and renewable energy consumption while lessening nonrenewable energy use. We found that control of corruption increases total, renewable, and nonrenewable energy consumption. Our findings showed that control of corruption moderates the effect of the net and market income inequality to increase total and nonrenewable energy consumption while reducing renewable energy consumption, with the marginal effect analysis suggesting that when control of corruption increases, income inequality (net) increases total energy and nonrenewable energy consumption while reducing renewable energy consumption. On the other hand, control of corruption was found to moderate the effect of income redistribution to reduce nonrenewable energy consumption while stimulating renewable energy consumption with the marginal effect suggesting that when control of corruption increases, both relative and absolute income redistribution reduces nonrenewable energy consumption while increasing renewable energy consumption. Our results further indicated that corruption, income inequality, and redistribution account for the variations in (non) renewable energy consumption across geographical regions.

These findings provide significant implications for strategies and policies for achieving net-zero and renewable energy transition. The Glasgow Climate Pact reemphasized the importance of keeping global warming at 1.5 °C and stressed that maintaining global warming at 1.5 °C requires phasing down nonrenewable energy and increasing renewable energy and energy efficiency. Our findings suggest that policies designed to achieve net-zero and renewable energy transition will fall short if income inequality and corruption are not addressed. Our study highlighted that addressing income inequality through income redistribution would contribute significantly to renewable energy consumption while reducing energy intensity and nonrenewable energy consumption. Policymakers could use redistributive policies such as taxation, especially taxing windfall profits from nonrenewable energy extraction, which would provide sufficient revenue to support any welfare program for poorer households to increase demand for renewable energy and enhance energy efficiency. In addition, another avenue for promoting energy transition and achieving energy efficiency is through control of corruption. Effectively managing corruption, especially in the energy sector, will eliminate inefficiency in energy policies and embezzlement and misuse of funds allocated for the generation of renewable. Making corruptive activities unpunishable would encourage moral hazards and can mar the success of policies, programs, and strategies for achieving the net-zero agenda through energy efficiency and renewable energy usage.

Another message from this study to policymakers is that the effect of income inequality and redistribution on renewable, nonrenewable, and total energy consumption depends on the institutional environment (control of corruption). For instance, it was demonstrated through marginal analysis that when corruption is effectively controlled, the pre-tax/transfer Gini index increases total energy, nonrenewable, and renewable energy consumption. The policy recommendation is that even when corruption is effectively managed, income inequality before tax will improve energy efficiency through renewable energy consumption but increase nonrenewable energy usage simultaneously. It was revealed through marginal analysis that when corruption is effectively controlled, the post-tax/transfer Gini index increases total and nonrenewable energy consumption while reducing renewable energy consumption. The policy recommendation is that when corruption control ensures that tax revenue is used to support energy transition policies and subsidize energy usage by poorer households, it would drive overall energy efficiency, reduce nonrenewable energy usage, and increase renewable energy consumption.

Further, it is demonstrated that with effective corruption control, relative and absolute income redistribution increases renewable energy while reducing nonrenewable energy consumption. Policymakers must effectively curtail corruption while strengthening redistributive policies to facilitate the transition towards a renewable energy economy. The policy relevance of this suggestion is that with control of corruption, both relative and absolute income redistribution would directly increase poor households' income. Increasing poorer household income enables these households to transition towards renewable energy relative to nonrenewable energy consumption. In other words, we argue that when income is redistributed directly to the households, it may increase their disposable income. Such income may be used to purchase renewable energy services relative to non-renewable energy. Therefore, we argue that when corruption control supports income redistribution to spur renewable energy transition, policymakers should use the tax revenue to support and subsidize renewable energy usage by poorer households. In summary, achieving the objectives of COP26 requires policymakers to improve institutional efficacy and income redistribution policies.

Although this study has made a modest empirical contribution to policy issues regarding achieving energy transition and net zero, it has some limitations. Our study relied on a global sample of 166 countries. Despite the advantage of using a large sample for valid generalization and conclusion, its policy recommendations may probably not apply to individual countries. This limitation allows future studies to use country-specific analytical approaches to investigate the impact of corruption, income inequality, and redistribution on energy transition. Given that the institutional environment matters in achieving the objectives of COP26, there is an avenue for future studies to explore the role of political ideologies in shaping energy transition in both developed and developing countries. This would contribute significantly to helping policymakers understand the political economy barriers in achieving energy transition and carbon neutrality in 2050. Nevertheless, our study provides policy insights that may guide global energy transition and net-zero policies.

CRediT authorship contribution statement

Alex O. Acheampong: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Elliot Boateng:** Conceptualization, Methodology, Formal analysis, Writing – original draft. **Collins Baah Annor:** Conceptualization, Methodology, Formal analysis, Writing – original draft.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors sincerely thank the handling Editor-in-Chief for the support. We are also grateful to the four anonymous reviewers for their valuable comments that helped improve this paper's quality. Nevertheless, the authors are responsible for all remaining errors.

Appendix

Table A1
List of countries included in the study.

Albania, Algeria, Angola, Argentina, Armenia, Australia, Austria, Azerbaijan, Bahamas, Bangladesh, Barbados, Belarus, Belgium, Belize, Benin, Bhutan, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cabo Verde, Cambodia, Cameroon, Central African Republic, Chad, Chile, China, Colombia, Comoros, Congo, Dem. Rep., Congo, Rep., Costa Rica, Cote d'Ivoire, Croatia, Cyprus, Czech Republic, Denmark, Djibouti, Dominica, Dominican Republic, Ecuador, Egypt, Arab Rep., El Salvador, Estonia, Eswatini, Ethiopia, Fiji, Finland, France, Gambia, The, Georgia, Germany, Ghana, Greece, Grenada, Guatemala, Guinea, Guinea-Bissau, Guyana, Haiti, Honduras, Hong Kong SAR, China, Hungary, Iceland, India, Indonesia, Iran, Islamic Rep., Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Korea, Dem. People's Rep., Kosovo, Kyrgyz Republic, Latvia, Lebanon, Lesotho, Liberia, Lithuania, Luxembourg, Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Micronesia, Fed. Sts., Moldova, Mongolia, Montenegro, Morocco, Mozambique, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, North Macedonia, Norway, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Puerto Rico, Qatar, Romania, Russian Federation, Rwanda, Samoa, Sao Tome and Principe, Senegal, Serbia, Seychelles, Sierra Leone, Singapore, Slovak Republic, Slovenia, Solomon Islands, South Africa, Spain, Sri Lanka, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, Sudan, Suriname, Sweden, Switzerland, Syrian Arab Republic, Tajikistan, Tanzania, Thailand, Timor-Leste, Togo, Tonga, Trinidad and Tobago, Tunisia, Turkey, Turkmenistan, Tuvalu, Uganda, Ukraine, United Kingdom, Uruguay, Uzbekistan, Venezuela, RB, Vietnam, Yemen, Rep., Zambia and Zimbabwe.

Table A2
Matrix of correlations.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) lnener	1.000													
(2) lnfosil	0.373	1.000												
(3) lnrenew	−0.638	−0.643	1.000											
(4) lngini_disp	−0.602	−0.143	0.458	1.000										
(5) lngini_mkt	−0.245	−0.092	0.407	0.658	1.000									
(6) lnrel_red	0.647	0.194	−0.335	−0.680	−0.031	1.000								
(7) lnabs_red	0.610	0.181	−0.278	−0.587	0.105	0.991	1.000							
(8) lnrgdpc	0.658	0.275	−0.212	−0.248	0.079	0.434	0.442	1.000						
(9) lnfdi	−0.001	−0.075	0.157	0.155	0.054	0.220	0.226	0.010	1.000					
(10) lntrade	0.250	−0.216	−0.092	−0.429	−0.230	0.211	0.178	0.001	0.369	1.000				
(11) lninfla	0.243	0.161	−0.266	−0.112	−0.203	0.040	0.012	0.035	−0.244	−0.124	1.000			
(12) lnoda	−0.078	0.211	−0.079	0.010	0.060	0.010	0.019	−0.393	0.038	−0.157	−0.054	1.000		
(13) lnsse	0.542	0.389	−0.388	−0.370	−0.032	0.579	0.572	0.569	0.142	0.031	−0.009	−0.062	1.000	
(14) corrup	0.345	0.051	0.074	−0.186	0.271	0.508	0.543	0.560	0.234	0.072	−0.217	−0.172	0.425	1.000

Table A3
Effect of income inequality, redistribution, and corruption on total energy consumption.

	Model 1 IV-GMM	Model 2	Model 3	Model 4	Model 5 Driscol-Kraay	Model 6	Model 7	Model 8	Model 9 FGLS	Model 10	Model 11	Model 12
lnrgdpc	0.652***	0.687***	0.546***	0.546***	0.648***	0.681***	0.542***	0.544***	0.520***	0.556***	0.483***	0.499***
	(0.030)	(0.031)	(0.038)	(0.039)	(0.029)	(0.030)	(0.044)	(0.045)	(0.015)	(0.013)	(0.025)	(0.026)
lnfdi	−0.009	−0.012	−0.094***	−0.095***	−0.004	−0.005	−0.094***	−0.095***	0.001	−0.003	−0.006	−0.007
	(0.021)	(0.021)	(0.022)	(0.022)	(0.011)	(0.011)	(0.022)	(0.021)	(0.004)	(0.004)	(0.007)	(0.007)
lntrade	0.260***	0.351***	0.443***	0.482***	0.250***	0.338***	0.451***	0.486***	0.037**	0.037*	0.141***	0.143***
	(0.046)	(0.046)	(0.051)	(0.053)	(0.020)	(0.027)	(0.018)	(0.023)	(0.018)	(0.019)	(0.030)	(0.031)
lninfla	0.126***	0.130***	0.136***	0.147***	0.120***	0.124***	0.139***	0.149***	0.012***	0.008**	0.020**	0.023***
	(0.016)	(0.017)	(0.021)	(0.022)	(0.010)	(0.014)	(0.040)	(0.042)	(0.004)	(0.003)	(0.008)	(0.008)
lnoda	0.084***	0.107***	0.105***	0.108***	0.085***	0.107***	0.106***	0.109***	0.000	0.000	0.003	0.006
	(0.015)	(0.015)	(0.019)	(0.019)	(0.021)	(0.027)	(0.027)	(0.030)	(0.005)	(0.004)	(0.009)	(0.010)
lnsse	0.161***	0.176***	0.147	0.229**	0.183***	0.201***	0.172***	0.246***	0.101***	0.114***	0.010	0.031
	(0.055)	(0.058)	(0.097)	(0.105)	(0.031)	(0.033)	(0.044)	(0.051)	(0.029)	(0.029)	(0.061)	(0.063)
corrup	−0.105***	−0.113***	−0.122***	−0.120***	−0.104***	−0.110***	−0.116***	−0.115***	−0.008	0.019	−0.052**	−0.049**
	(0.028)	(0.029)	(0.035)	(0.035)	(0.012)	(0.031)	(0.017)	(0.014)	(0.016)	(0.014)	(0.024)	(0.025)
lngini_disp	−0.803***				−0.820***				−1.074***			
	(0.078)				(0.065)				(0.053)			
lngini_mkt		−0.403***				−0.450***				−0.697***		
		(0.127)				(0.083)				(0.078)		
lnrel_red			0.256***				0.245***				0.250***	
			(0.020)				(0.024)				(0.018)	
lnabs_red				0.223***				0.216***				0.229***
				(0.020)				(0.029)				(0.018)
Constant	0.792	−1.821**	−2.806***	−3.173***	0.826	−1.640**	−2.926***	−3.258***	5.926***	4.299***	1.742***	1.661***
	(0.683)	(0.712)	(0.673)	(0.699)	(0.781)	(0.735)	(0.892)	(1.000)	(0.292)	(0.328)	(0.355)	(0.370)
Observations	849	849	397	397	863	863	398	398	855	855	398	398
R2	0.738	0.709	0.731	0.714	0.735	0.707	0.731	0.715				
F-statistics	264,172.535	261,192.689	5279.274	5825.632								

Standard errors in parentheses.
* $p < 0.10$,.
** $p < 0.05$,.
*** $p < 0.01$.

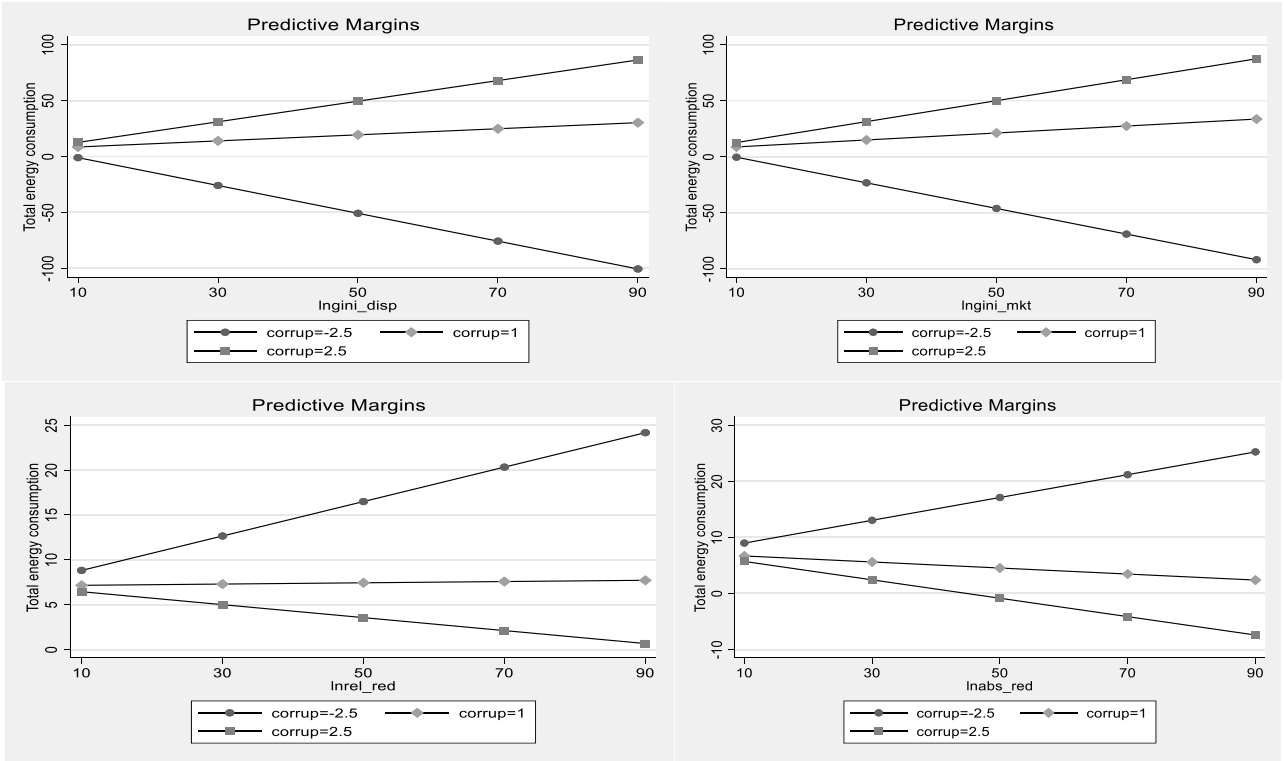

Fig. A1. Predictive margins results with total energy consumption as the dependant variable.

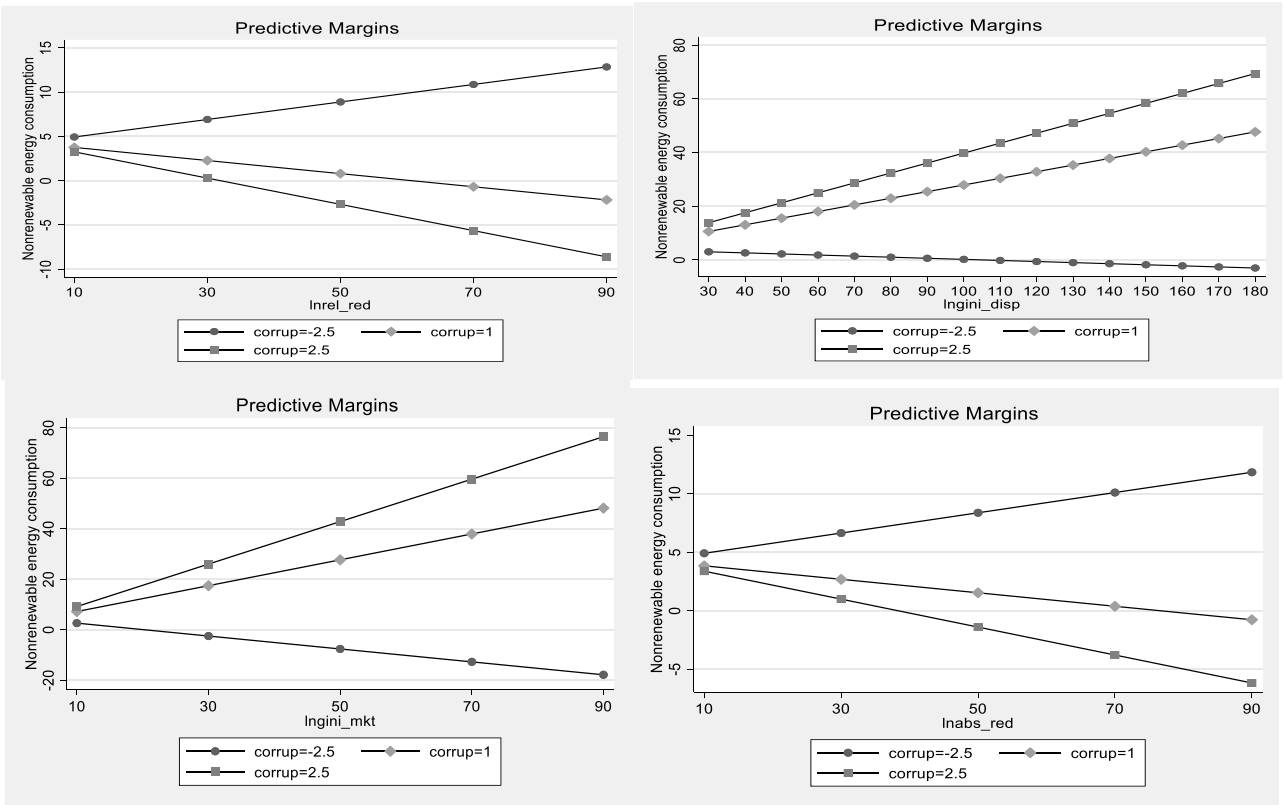

Fig. A2. Predictive margins results with nonrenewable energy consumption as the dependant variable.

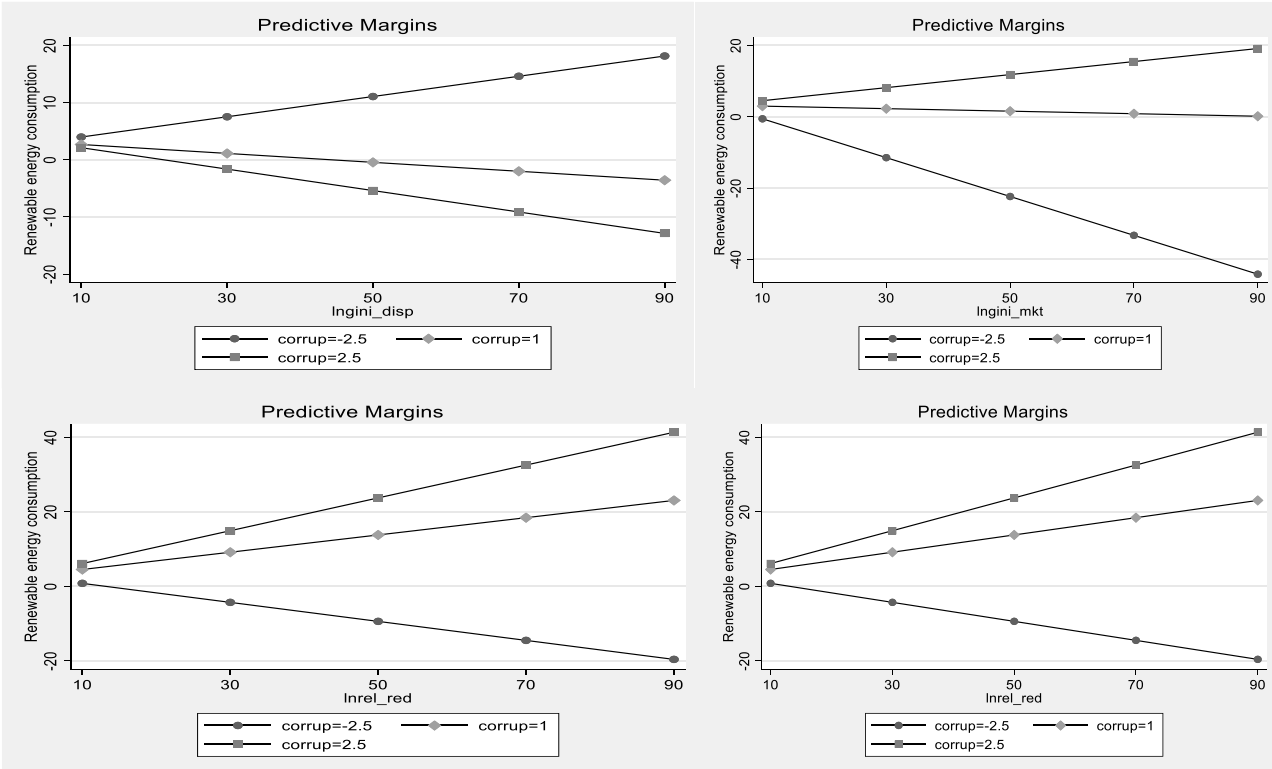

Fig. A3. Predictive margins results with renewable energy consumption as the dependant variable.

References

Acheampong, A.O., Boateng, E., Amponsah, M., Dzator, J., 2021a. Revisiting the economic growth–energy consumption nexus: does globalization matter? Energy Econ. 102, 105472 https://doi.org/10.1016/j.eneco.2021.105472.

Acheampong, A.O., Dzator, J., Savage, D.A., 2021b. Renewable energy, CO2 emissions and economic growth in sub-Saharan Africa: does institutional quality matter? J. Policy Model. 43 (5), 1070–1093. https://doi.org/10.1016/j.jpolmod.2021.03.011.

Adams, S., Klobodu, E.K.M., 2016. Financial development, control of corruption and income inequality. Int. Rev. Appl. Econ. 30 (6), 790–808.

Akhbari, R., Nejati, M., 2019. Does the effect of corruption on carbon emissions vary in different countries? Environ. Sci. (Ruse) 17 (1), 105–120.

Alesina, A., Rodrik, D., 1994. Distributive politics and economic growth. Q. J. Econ. 109 (2), 465–490. https://doi.org/10.2307/2118470.

Amuakwa-Mensah, F., Klege, R.A., Adom, P.K., Amoah, A., Hagan, E., 2018. Unveiling the energy saving role of banking performance in Sub-Sahara Africa. Energy Econ. 74, 828–842. https://doi.org/10.1016/j.eneco.2018.07.031.

Anton, S.G., Afloarei Nucu, A.E., 2020. The effect of financial development on renewable energy consumption. A panel data approach. Renew. Energy 147, 330–338. https://doi.org/10.1016/j.renene.2019.09.005.

Arellano, M., Bond, S., 1991. Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. Rev. Econ. Stud. 58 (2), 277–297.

Asongu, S.A., Odhiambo, N.M., 2021. Inequality, finance and renewable energy consumption in Sub-Saharan Africa. Renew. Energy 165, 678–688. https://doi.org/10.1016/j.renene.2020.11.062.

Asongu, S.A., Iheonu, C.O., Odo, K.O., 2019. The conditional relationship between renewable energy and environmental quality in sub-Saharan Africa. Environ. Sci. Pollut. Res. Int. 26 (36), 36993–37000.

Awaworyi Churchill, S., Ivanovski, K., Munyanyi, M.E., 2021. Income inequality and renewable energy consumption: time-varying non-parametric evidence. J. Clean. Prod. 296, 126306 https://doi.org/10.1016/j.jclepro.2021.126306.

Bai, C., Feng, C., Yan, H., Yi, X., Chen, Z., Wei, W., 2020. Will income inequality influence the abatement effect of renewable energy technological innovation on carbon dioxide emissions? J. Environ. Manage. 264, 110482.

Bakre, O., Lauwo, S.G., McCartney, S., 2017. Western accounting reforms and accountability in wealth redistribution in patronage-based Nigerian society. Account. Audit. Accountabil. J.

Barro, R.J., 2000. Inequality and growth in a panel of countries. J. Econ. Growth 5 (1), 5–32.

Bataille, C., Melton, N., 2017. Energy efficiency and economic growth: a retrospective CGE analysis for Canada from 2002 to 2012. Energy Econ. 64, 118–130. https://doi.org/10.1016/j.eneco.2017.03.008.

Benfratello, L., Del Monte, A., Pennacchio, L., 2018. Corruption and public debt: a cross-country analysis. Appl. Econ. Lett. 25 (5), 340–344.

Berisha, E., Meszaros, J., Olson, E., 2018. Income inequality, equities, household debt, and interest rates: evidence from a century of data. J. Int. Money Finance 80, 1–14.

Berthe, A., Elie, L., 2015. Mechanisms explaining the impact of economic inequality on environmental deterioration. Ecol. Econ. 116, 191–200.

Bhattacharya, M., Awaworyi Churchill, S., Paramati, S.R, 2017. The dynamic impact of renewable energy and institutions on economic output and CO2 emissions across regions. Renew. Energy 111, 157–167. https://doi.org/10.1016/j.renene.2017.03.102.

Bhorat, H., Naidoo, K., Odusola, A., Cornia, G., Conceição, P., 2017. Income Inequality Trends in Sub-Saharan Africa: Divergence, determinants, and Consequences. UNDP Regional Bureau for Africa, New York, NY.

Biswas, A.K., Farzanegan, M.R., Thum, M., 2012. Pollution, shadow economy and corruption: theory and evidence. Ecol. Econ. 75, 114–125.

Blau, J., 2017. The Paris Agreement: Climate change, solidarity, and Human Rights. Springer.

Blundell, R., Bond, S., 1998. Initial conditions and moment restrictions in dynamic panel data models. J. Econom. 87 (1), 115–143. https://doi.org/10.1016/S0304-4076(98)00009-8.

Borghesi, S., 2006. Income inequality and the environmental Kuznets curve. Environ. Inequality Collective Action 33.

Boyce, J.K., 1994. Inequality as a cause of environmental degradation. Ecol. Econ. 11 (3), 169–178. https://doi.org/10.1016/0921-8009(94)90198-8.

Boyce, J.K., 2007. Inequality and environment. Inequality Cooper. Environ. Sustain. 314.

Boyle, A., 2018. Climate change, the Paris Agreement and human rights. Int. Compar. Law Quarterly 67 (4), 759–777.

Busso, M.M., Julián. (2020). The inequality crisis: latin America and the Caribbean at the crossroads.

Brambor, T., Clark, W.R., Golder, M., 2006. Understanding interaction models: improving empirical analyses. Political Anal. 14 (1), 63–82.

Cadoret, I., Padovano, F., 2016. The political drivers of renewable energies policies. Energy Econ. 56, 261–269.

Chancel, L., 2022. Global carbon inequality over 1990–2019. Nature Sustain. 1–8.

Chancel, L., Piketty, T., Saez, E., Zucman, G., 2022. World Inequality Report 2022. Harvard University Press.

Chong, A., Calderon, C., 2000. Institutional quality and poverty measures in a cross-section of countries. Econ. Governance 1 (2), 123–135.

Churchill, S.A., Ivanovski, K., Munyanyi, M.E., 2021. Income inequality and renewable energy consumption: time-varying non-parametric evidence. J. Clean. Prod. 296, 126306.
Cole, M.A., 2006. Does trade liberalization increase national energy use? Econ. Lett. 92 (1), 108–112. https://doi.org/10.1016/j.econlet.2006.01.018.
Del Mar Salinas-Jiménez, M., Salinas-Jiménez, J., 2007. Corruption, efficiency and productivity in OECD countries. J. Policy Model. 29 (6), 903–915.
Dogan, E., Seker, F., 2016. Determinants of CO2 emissions in the European Union: the role of renewable and nonrenewable energy. Renew. Energy 94, 429–439.
Dogan, E., Seker, F., 2016a. Determinants of CO2 emissions in the European Union: the role of renewable and non-renewable energy. Renew. Energy 94, 429–439. https://doi.org/10.1016/j.renene.2016.03.078.
Dogan, E., Seker, F., 2016b. The influence of real output, renewable and non-renewable energy, trade and financial development on carbon emissions in the top renewable energy countries. Renewable Sustainable Energy Rev. 60, 1074–1085. https://doi.org/10.1016/j.rser.2016.02.006.
Dogan, E., Chishti, M.Z., Karimi Alavijeh, N., Tzeremes, P., 2022. The roles of technology and Kyoto Protocol in energy transition towards COP26 targets: evidence from the novel GMM-PVAR approach for G-7 countries. Technol. Forecast. Soc. Change 181, 121756. https://doi.org/10.1016/j.techfore.2022.121756.
Doytch, N., Narayan, S., 2016. Does FDI influence renewable energy consumption? An analysis of sectoral FDI impact on renewable and non-renewable industrial energy consumption. Energy Econ. 54, 291–301. https://doi.org/10.1016/j.eneco.2015.12.010.
Fisman, R., Svensson, J., 2007. Are corruption and taxation really harmful to growth? Firm level evidence. J. Dev. Econ. 83 (1), 63–75.
Fredriksson, P.G., Vollebergh, H.R., Dijkgraaf, E., 2004. Corruption and energy efficiency in OECD countries: theory and evidence. J. Environ. Econ. Manage. 47 (2), 207–231.
Gennaioli, C., Tavoni, M., 2016. Clean or dirty energy: evidence of corruption in the renewable energy sector. Public Choice 166 (3), 261–290.
Glaeser, E., Scheinkman, J., Shleifer, A., 2003. The injustice of inequality. J. Monet. Econ. 50 (1), 199–222. https://doi.org/10.1016/s0304-3932(02)00204-0.
Goñi, E., Humberto López, J., Servén, L., 2011. Fiscal redistribution and income inequality in Latin America. World Dev. 39 (9), 1558–1569. https://doi.org/10.1016/j.worlddev.2011.04.025.
Grunewald, N., Klasen, S., Martínez-Zarzoso, I., Muris, C., 2017. The trade-off between income inequality and carbon dioxide emissions. Ecol. Econ. 142, 249–256.
Güney, T., 2019. Renewable energy, nonrenewable energy and sustainable development. Int. J. Sustainable Dev. World Ecol. 26 (5), 389–397.
Gupta, S., Davoodi, H., Alonso-Terme, R., 2002. Does corruption affect income inequality and poverty? Econ. Governance 3 (1), 23–45. https://doi.org/10.1007/s101010100039.
Gustavson, M., Sundström, A., 2018. Organizing the audit society: does good auditing generate less public sector corruption? Adm. Soc. 50 (10), 1508–1532.
Hao, Y., Chen, H., Zhang, Q., 2016. Will income inequality affect environmental quality? Analysis based on China's provincial panel data. Ecol. Indic. 67, 533–542.
Heerink, N., Mulatu, A., Bulte, E., 2001. Income inequality and the environment: aggregation bias in environmental Kuznets curves. Ecol. Econ. 38 (3), 359–367.
IEA, 2022. Africa Energy Outlook 2022. IEA, Paris. https://www.iea.org/reports/africa-energy-outlook-2022. License: CC BY 4.0.
Inglesi-Lotz, R., Dogan, E., 2018. The role of renewable versus non-renewable energy to the level of CO2 emissions a panel analysis of sub- Saharan Africa's Big 10 electricity generators. Renew. Energy 123, 36–43. https://doi.org/10.1016/j.renene.2018.02.041.
Jin, T., Kim, J., 2018. What is better for mitigating carbon emissions–renewable energy or nuclear energy? A panel data analysis. Renewable Sustainable Energy Rev. 91, 464–471.
Johansson, M., Gustafsson, B., 1997. In Search for a Smoking Gun: What Makes Income Inequality Vary Over Time in Different Countries? (No. 172). LIS Cross-National Data Center in Luxembourg.
Kaasa, A., 2003. Factors influencing income inequality in transition economies. University of Tartu Economics and Business Administration Working Paper Series.
Kaufmann, D., Kraay, A., Mastruzzi, M., 2011. The worldwide governance indicators: methodology and analytical issues. Hague J. Rule Law 3 (2), 220–246.
Keneck-Massil, J., Nomo-Beyala, C., Owoundi, F., 2021. The corruption and income inequality puzzle: does political power distribution matter? Econ. Model. 103, 105610 https://doi.org/10.1016/j.econmod.2021.105610.
Kidd, S., Athias, D., Nastasi, S., & Pop, A. (2022). Inequality and social security in the Asia-Pacific region: united Nations Development Programme.
Kyophilavong, P., Shahbaz, M., Anwar, S., Masood, S., 2015. The energy-growth nexus in Thailand: does trade openness boost up energy consumption? Renewable Sustainable Energy Rev. 46, 265–274. https://doi.org/10.1016/j.rser.2015.02.004.
Laurent, E., 2015. Social-Ecology: exploring the missing link in sustainable development. HAL Id:hal-01136326. https://hal-sciencespo.archives-ouvertes.fr/hal01136326.
Leitão, A., 2010. Corruption and the environmental Kuznets curve: empirical evidence for sulfur. Ecol. Econ. 69 (11), 2191–2201.
Li, H., Xu, L.C., Zou, H.-f., 2000. Corruption, income distribution, and growth. Econ. Politics 12 (2), 155–182. https://doi.org/10.1111/1468-0343.00073.
Lisciandra, M., Migliardo, C., 2017. An empirical study of the impact of corruption on environmental performance: evidence from panel data. Environ. Resource Econ. 68 (2), 297–318.
Liu, L., Zhou, C., Huang, J., Hao, Y., 2018. The impact of financial development on energy demand: evidence from China. Emerg. Mark. Finance Trade 54 (2), 269–287. https://doi.org/10.1080/1540496X.2017.1358609.
Liu, Z., Zhang, H., Zhang, Y.J., Qin, C.X., 2020. How does income inequality affect energy efficiency? Empirical evidence from 33 Belt and Road Initiative countries. J. Clean. Prod. 269, 122421.
Lopez, R., Mitra, S., 2000. Corruption, pollution, and the Kuznets environment curve. J. Environ. Econ. Manage. 40 (2), 137–150.
Martinot, E., Chaurey, A., Lew, D., Moreira, J.R., Wamukonya, N., 2002. Renewable energy markets in developing countries. Annu. Rev. Energy Env. 27 (1), 309–348.
Masih, A.M., Masih, R., 1996. Energy consumption, real income and temporal causality: results from a multi-country study based on cointegration and error-correction modelling techniques. Energy Econ. 18 (3), 165–183.
Mehrara, M., Rezaei, S., Razi, D.H., 2015. Determinants of renewable energy consumption among ECO countries; based on Bayesian model averaging and weighted-average least square. Int. Lett. Social Humanistic Sci. 54, 96–109.
Milanovic, B., 2000. The median-voter hypothesis, income inequality, and income redistribution: an empirical test with the required data. Eur. J. Polit. Econ. 16 (3), 367–410. https://doi.org/10.1016/S0176-2680(00)00014-8.
Mountford, H., Waskow, D., Gonzalez, L., Gajjar, C., Cogswell, N., Holt, M., ... Gerholdt, R. (2021). COP26: key outcomes from the UN climate talks in Glasgow. Retrieved from https://www.wri.org/insights/cop26-key-outcomes-un-climate-talks-glasgow?utm_medium=social&utm_source=twitter&utm_campaign=socialmedia.
Murshed, M., Alam, M.S., 2021. Estimating the macroeconomic determinants of total, renewable, and non-renewable energy demands in Bangladesh: the role of technological innovations. Environ. Sci. Pollut. Res. Int. 28 (23), 30176–30196. https://doi.org/10.1007/s11356-021-12516-6.
Nair, K., Shadman, S., Chin, C.M., Sakundarini, N., Yap, E.H., Koyande, A., 2021. Developing a system dynamics model to study the impact of renewable energy in the short-and long-term energy security. Mater. Sci. Energy Technol. 4, 391–397.
Njoh, A.J., Etta, S., Essia, U., Ngyah-Etchutambe, I., Enomah, L.E., Tabrey, H.T., Tarke, M.O., 2019. Implications of institutional frameworks for renewable energy policy administration: case study of the Esaghem, Cameroon community PV solar electrification project. Energy Policy 128, 17–24.
Omri, A., Nguyen, D.K., 2014. On the determinants of renewable energy consumption: international evidence. Energy 72, 554–560. https://doi.org/10.1016/j.energy.2014.05.081.
Opeyemi, A., Uchenna, E., Simplice, A., Evans, O., 2019. Renewable energy, trade performance and the conditional role of finance and institutional capacity in sub-Sahara African countries. Energy Policy 132, 490–498.
Paramati, S.R., Apergis, N., Ummalla, M., 2017. Financing clean energy projects through domestic and foreign capital: the role of political cooperation among the EU, the G20 and OECD countries. Energy Econ. 61, 62–71. https://doi.org/10.1016/j.eneco.2016.11.001.
Paramati, S.R., Ummalla, M., Apergis, N., 2016. The effect of foreign direct investment and stock market growth on clean energy use across a panel of emerging market economies. Energy Econ. 56, 29–41. https://doi.org/10.1016/j.eneco.2016.02.008.
Policardo, L., Carrera, E.J.S., Risso, W.A., 2019. Causality between income inequality and corruption in OECD countries. World Dev. Perspect. 14, 100102.
Pörtner, H.O., Roberts, D.C., Adams, H., Adler, C., Aldunce, P., Ali, E., Biesbroek, R., 2022. Climate change 2022: impacts, adaptation and vulnerability. IPCC Sixth Assessment Report. Retrieved from. https://www.ipcc.ch/report/ar6/wg2/.
Rafiq, S., Salim, R., Nielsen, I., 2016. Urbanization, openness, emissions, and energy intensity: a study of increasingly urbanized emerging economies. Energy Econ. 56, 20–28. https://doi.org/10.1016/j.eneco.2016.02.007.
Sadorsky, P., 2009. Renewable energy consumption, CO2 emissions and oil prices in the G7 countries. Energy Econ. 31 (3), 456–462. https://doi.org/10.1016/j.eneco.2008.12.010.
Sadorsky, P., 2010. The impact of financial development on energy consumption in emerging economies. Energy Policy 38 (5), 2528–2535. https://doi.org/10.1016/j.enpol.2009.12.048.
Sadorsky, P., 2011a. Financial development and energy consumption in Central and Eastern European frontier economies. Energy Policy 39 (2), 999–1006. https://doi.org/10.1016/j.enpol.2010.11.034.
Sadorsky, P., 2011b. Trade and energy consumption in the Middle East. Energy Econ. 33 (5), 739–749. https://doi.org/10.1016/j.eneco.2010.12.012.
Sadorsky, P., 2012. Energy consumption, output and trade in South America. Energy Econ. 34 (2), 476–488. https://doi.org/10.1016/j.eneco.2011.12.008.
Sampedro, J., Iyer, G., Msangi, S., Waldhoff, S., Hejazi, M., Edmonds, J.A., 2022. Implications of different income distributions for future residential energy demand in the US. Environ. Res. Lett. 17 (1), 014031.
Sari, R., Soytas, U., 2007. The growth of income and energy consumption in six developing countries. Energy Policy 35 (2), 889–898.
Sbia, R., Shahbaz, M., Hamdi, H., 2014. A contribution of foreign direct investment, clean energy, trade openness, carbon emissions and economic growth to energy demand in UAE. Econ. Model. 36, 191–197. https://doi.org/10.1016/j.econmod.2013.09.047.
Scruggs, L.A., 1998. Political and economic inequality and the environment. Ecol. Econ. 26 (3), 259–275.
Sequeira, T.N., Santos, M.S., 2018. Renewable energy and politics: a systematic review and new evidence. J. Clean. Prod. 192, 553–568. https://doi.org/10.1016/j.jclepro.2018.04.190.
Shahbaz, M., Rizvi, S.K.A., Dong, K., Vo, X.V., 2022. Fiscal decentralization as new determinant of renewable energy demand in China: the role of income inequality and urbanization. Renew. Energy 187, 68–80.
Sharma, S.S., 2011. Determinants of carbon dioxide emissions: empirical evidence from 69 countries. Appl. Energy 88 (1), 376–382.
Solt, F., 2016. The standardized world income inequality database. Soc. Sci. Q. 97 (5), 1267–1281.

Soytas, U., Sari, R., 2003. Energy consumption and GDP: causality relationship in G-7 countries and emerging markets. Energy Econ. 25 (1), 33–37. https://doi.org/10.1016/S0140-9883(02)00009-9.

Soytas, U., Sari, R., 2009. Energy consumption, economic growth, and carbon emissions: challenges faced by an EU candidate member. Ecol. Econ. 68 (6), 1667–1675. https://doi.org/10.1016/j.ecolecon.2007.06.014.

Strunz, S., Gawel, E., Lehmann, P., 2016. The political economy of renewable energy policies in Germany and the EU. Util. Policy 42, 33–41.

Sulemana, I., Kpienbaareh, D., 2018. An empirical examination of the relationship between income inequality and corruption in Africa. Econ. Anal. Policy 60, 27–42.

Tan, Y., Uprasen, U., 2021. Carbon neutrality potential of the ASEAN-5 countries: implications from asymmetric effects of income inequality on renewable energy consumption. J. Environ. Manage. 299, 113635.

Thiam, D.R., 2011. Renewable energy, poverty alleviation and developing nations: evidence from Senegal. J. Energy Southern Africa 22 (3), 23–34.

Troster, V., Shahbaz, M., Uddin, G.S., 2018. Renewable energy, oil prices, and economic activity: a Granger-causality in quantiles analysis. Energy Econ. 70, 440–452. https://doi.org/10.1016/j.eneco.2018.01.029.

Uzar, U., 2020. Is income inequality a driver for renewable energy consumption? J. Clean. Prod. 255, 120287 https://doi.org/10.1016/j.jclepro.2020.120287.

Uzar, U., Eyuboglu, K., 2019. The nexus between income inequality and CO2 emissions in Turkey. J. Clean. Prod. 227, 149–157.

Villanthenkodath, M.A., Mahalik, M.K., Patel, G., 2023. Effects of foreign aid and energy aid inflows on renewable and non-renewable electricity production in BRICS countries. Environ. Sci. Pollut. Res. Int. 30 (3), 7236–7255. https://doi.org/10.1007/s11356-022-22730-5.

Wang, L., Vo, X.V., Shahbaz, M., Ak, A., 2020. Globalization and carbon emissions: is there any role of agriculture value-added, financial development, and natural resource rent in the aftermath of COP21? J. Environ. Manage. 268, 110712.

Wilkinson, R., Pickett, K., 2010. The Spirit level: Why equality is Better For Everyone. Penguin, UK.

Wolde-Rufael, Y., Idowu, S., 2017. Income distribution and CO2 emission: a comparative analysis for China and India. Renewable Sustainable Energy Rev. 74, 1336–1345.

Xu, D., Sheraz, M., Hassan, A., Sinha, A., Ullah, S., 2022. Financial development, renewable energy and CO2 emission in G7 countries: new evidence from non-linear and asymmetric analysis. Energy Econ. 109, 105994 https://doi.org/10.1016/j.eneco.2022.105994.

Yang, Z., Zhang, M., Liu, L., Zhou, D., 2022. Can renewable energy investment reduce carbon dioxide emissions? Evidence from scale and structure. Energy Econ. 112, 106181 https://doi.org/10.1016/j.eneco.2022.106181.

Yao, Y., Ivanovski, K., Inekwe, J., Smyth, R., 2019. Human capital and energy consumption: evidence from OECD countries. Energy Econ. 84, 104534 https://doi.org/10.1016/j.eneco.2019.104534.

Zhang, M., Zhang, S., Lee, C.C., Zhou, D., 2021. Effects of trade openness on renewable energy consumption in OECD countries: new insights from panel smooth transition regression modelling. Energy Econ. 104, 105649 https://doi.org/10.1016/j.eneco.2021.105649.

Zhao, P., Lu, Z., Fang, J., Paramati, S.R., Jiang, K., 2020. Determinants of renewable and non-renewable energy demand in China. Struct. Change Econ. Dyn. 54, 202–209. https://doi.org/10.1016/j.strueco.2020.05.002.